LEVERAGED DEFI LONG–SHORT POSITIONS UNDER LIQUIDATION AND JUMP RISK: A SEMI-ANALYTIC EVALUATION FRAMEWORK

We develop a semi-analytic framework for evaluating leveraged DeFi long–short positions whose solvency is governed by a protocol health factor. Under bivariate Kou jump–diffusion dynamics, an exponential change of measure absorbs the short-leg price and reduces every admissible payoff moment to a killed expectation of the log collateral-to-liability ratio, whose Laplace-domain resolvent is a 3×3 linear system for every moment order. The same barrier system delivers four further objects: the exact liquidation recovery under jump overshoot, through bivariate tilts; the law of the overshoot itself — a creeping atom and two exponential phases whose weights follow from three hitting moments — and with it the exact probability of bad debt; the expected logarithm of terminal wealth, so that Kelly sizing is semi-analytic rather than simulated; and deterministic and Markov-modulated interest-rate carry, the latter through a 6×6 resolvent. The resulting objective-independent *evaluation map* from initial buffer to liquidation probability, payoff moments and recovery is validated against bridge-corrected Monte Carlo and is two to three orders of magnitude faster than simulation. On 3.7 years of Aave v3 Ethereum data, the map's one-month liquidation probabilities lie above realised frequencies out of sample, systematically so for some predictors, deterministic carry is small in barrier units for cross-asset pairs and up to a barrier standard deviation for some stablecoin pairs, and the jump-overshoot correction ranges from -3% to -11% depending on the calibration window. Applied to sizing, the map gives a closed-form break-even drift above which leverage pays, and the data cannot settle whether a pair's drift exceeds it. A model-free replay of 37 Aave v3 pairs under the protocol's liquidation rules shows that leverage rarely paid ex post: at $1.25\times$ it beat the unlevered position in 4 pairs, never significantly, and at $2\times$ over 2025 it lost in every cross-asset pair. The same replay identifies the model's main restriction. With independent jumps, the calibration cannot reproduce the legs' common jumps, so it understates their covariance, overstates spread variance and, for WBTC/WETH, turns a negative realised break-even positive. Common jumps are the natural next step.

Keywords: DeFi lending; leveraged long–short; health factor; Kou double-exponential jump diffusion; first-passage problem; Laplace resolvent; jump overshoot; bad debt; Kelly criterion; interest-rate carry; regime switching

JEL Classification: G11; G12; C65

1. Introduction

DeFi lending protocols such as Aave and Compound enable a leveraged long–short strategy: deposit a cryptocurrency as collateral, borrow a second asset, and sell it short. The position is governed by a real-time *health factor* — a borrowing-power-adjusted ratio of collateral to liability — and is liquidated automatically the instant that ratio falls below one. The practitioner must choose the initial holdings to balance two forces: a more aggressive position amplifies returns on surviving paths but raises the probability of forced exit. Crypto spreads are fat-tailed, so realistic liquidation probabilities require jump dynamics.

We develop a semi-analytic evaluation map for this sizing problem under the Kou double-exponential jump–diffusion (Kou, 2002), with the liquidation boundary treated as absorbing. The solution rests on two reductions. First, an exponential change of measure absorbs the short-leg price factor into the path-measure density, collapsing the bivariate barrier problem to a univariate first-exit problem for the log-ratio process under a tilted Lévy measure. Second, because exponentials are eigenfunctions of the Kou generator, the Laplace-domain resolvent reduces to a 3×3 linear system. The system stays 3×3 for every admissible moment order k (those with $k \max(\eta_{1,+}, \eta_{2,+}) < 1$): changing k changes the forcing polynomial and the phase rates, not the number of decaying modes. Time-domain quantities follow by Talbot inversion (Talbot, 1979). The whole problem becomes a smooth scalar search over the *initial log health factor* h_0, with noise-free outputs that are two to three orders of magnitude faster than Monte Carlo, whether naive or accuracy-matched (Section 6.1).

The paper sits between the DeFi liquidation literature and the first-passage literature for jump diffusions. Qin et al. (2021) and Heimbach and Huang (2024) document the incentive structure and leverage amplification of DeFi liquidations empirically, without addressing ex-ante sizing. Kou and Wang (2003) derive Laplace-transform first-passage results under double-exponential jumps, and Sepp (2004) reduces barrier option pricing to a finite algebraic system by the mechanism used here; the finiteness is generic for exponential-type Lévy models — hyperexponential (Cai, 2009), mixed-exponential (Cai and Kou, 2011) and phase-type (Asmussen et al., 2004) — with dimension set by the number of exponential phases. We borrow this machinery at its core. What is new is what it is made to compute: the payoff is a barrier-killed long–short *ratio*, which the change-of-numeraire tilt brings within reach of the machinery, and bivariate tilts of the same barrier system then deliver the joint law of the leg values at liquidation, the overshoot law and the expected log of wealth. More distantly, Jacobs et al. (1999) advocate joint optimisation of long and short legs and Dang and Forsyth (2016) study dynamic mean–variance control under jump risk; our setting is static and the binding constraint is an endogenous first-passage event.

The paper makes five contributions. *(i)* It reduces the position to the scalar initial log health factor h_0 and assembles an objective-independent *health-buffer evaluation map* (7): the liquidation probability and every admissible raw killed moment $\mathbb{E}[\Pi_T^k]$, from one 3×3 barrier system (Sections 2–5). *(ii)* It computes the exact liquidation recovery under jump overshoot from joint hitting moments under bivariate tilts, and shows that the overshoot is a creeping atom plus two exponential phases whose weights follow from three hitting moments. This gives the exact bad-debt probability (Section 7, Proposition 7.1). *(iii)* It evaluates the expected log of terminal wealth semi-analytically, by a convergent series of tilted killed moments on survival and the overshoot law on liquidation, so that Kelly sizing needs no simulation (Section 7.5). *(iv)* It incorporates protocol interest: deterministic carry as a per-leg drift shift (Proposition 8.1) and the convexity of the kinked rate curve through a two-state Markov-modulated carry with a 6×6 resolvent (Proposition C.1). *(v)* It validates the map numerically — against an accuracy-matched, Brownian-bridge-corrected simulator (Section 6.1) — and empirically on 3.7 years of Aave v3 data, where an out-of-sample backtest over 33 months shows its one-month liquidation probabilities to lie above realised frequencies, significantly so for some predictors (Appendix A.3). Section 9 then applies the map to sizing. Leverage pays once the relative-drift view exceeds a closed-form break-even of a few percent per year, and the data cannot establish whether it does. A model-free replay of 37 Aave v3 pairs with the protocol's liquidation rules (Section 9.3) shows that leverage rarely paid ex post, and identifies the model's independent-jump restriction as a main reason why its break-evens and liquidation probabilities differ from the realised ones. A decision-theoretic treatment of the sizing problem is outlined as an extension.

The framework abstracts from liquidation execution, dynamic collateral management and oracle delays, and from protocol features that rescale the effective collateral factor or cap position size (E-mode, supply/borrow caps, isolation mode); E-mode would enter only through a larger b.

Section 2 sets up the position and the scalar reduction; Section 3 specifies the dynamics and tilt;

Sections 4–5 derive the moments and resolvents; Section 6 reports numerical results; Section 7 adds recovery, the overshoot law and expected log wealth; Section 8 adds carry; Section 9 applies the map to sizing, confronts it with a realised backtest and outlines the decision-theoretic extension; Section 10 concludes. The appendices give the calibration and its full-history validation, the pair cross-section, the 6×6 resolvent, and the proofs.

2. Problem Setup

2.1. Surviving-path terminal wealth

Let $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space carrying two strictly positive prices $S_{i,t} = S_{i,0}e^{X_{i,t}}$, $X_{i,0} = 0$, $i = 1,2$, with log-price processes specified in Section 3. The investor holds $n_1 > 0$ units of S_1 (collateral, long) and owes $n_2 > 0$ units of S_2 (debt, short), with leg values $V_{1,t} := n_1 S_{1,t}$ and $V_{2,t} := n_2 S_{2,t}$. Fix $b \in (0,1)$, the Aave v3 *liquidation threshold* of S_1 (distinct from the lower maximum LTV at origination). The protocol liquidates the first time $bV_{1,t} \leq V_{2,t}$, i.e. when the *health factor*

$$H_t := \frac{bV_{1,t}}{V_{2,t}} = \frac{bn_1 S_{1,t}}{n_2 S_{2,t}}, \qquad h_t := \log H_t, \tag{1}$$

falls below one. Writing

$$h_0 := \log\Big(\frac{bn_1 S_{1,0}}{n_2 S_{2,0}}\Big), \qquad Z_t := X_{1,t} - X_{2,t},$$

we have $h_t = h_0 + Z_t$, and the liquidation time is

$$\tau := \inf\{t \geq 0 :\ h_t < 0\} = \inf\{t \geq 0 :\ Z_t < -h_0\}.$$

The one-dimensional process Z carries all the liquidation risk, and h_0 is the initial log-distance of the health factor from the boundary $H = 1$.

The *surviving-path terminal wealth* at horizon T is

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T})\,\mathbf{1}_{\{\tau > T\}}.$$

On $\{\tau > T\}$, $H_T \geq 1$ gives $n_1 S_{1,T} \geq n_2 S_{2,T}/b > n_2 S_{2,T}$, so equity is strictly positive on every surviving path and no positive-part truncation is needed. Liquidated paths are assigned zero; Remark 1 explains the convention and Section 7 restores the recovery value.

Remark 1 (Liquidation convention). On Aave v3 a keeper may repay up to 50% of the debt (100% below a secondary threshold) and claims a liquidation bonus of 5–15% of seized collateral; Compound is similar (Qin et al., 2021; Heimbach and Huang, 2024). Execution is settled on-chain, so the borrower's residual depends on pool liquidity, gas prices and keeper speed. We treat the barrier as absorbing and set $\Pi_T := 0$ on $\{\tau \leq T\}$. Two caveats. First, a partial liquidation typically *restores* $H > 1$ and the position continues at reduced size, so absorption models a borrower who is fully closed out or treats the first liquidation as terminal. Second, the zero is a bookkeeping convention that moves recovery out of Π_T, not a claim that it is worthless. It is *not* conservative for survival-conditional moments — conditioning on $\{\tau > T\}$ discards exactly the liquidated paths — which is one reason Section 2.3 builds the objective on unconditional wealth. What it does bound is unconditional wealth on the loss event: it never overstates W_T relative to any contract with non-negative recovery.

Remark 2 (Static position). A continuously monitored book could top up collateral or unwind the short leg, replacing the absorbing barrier with a controlled one. We do not pursue this. During a flash crash — the jump-driven event that dominates liquidation — a position may be liquidated within one or two

blocks, before a remedial transaction confirms; and the ex-ante buffer decision is cleanly isolated by a static model. The static model is therefore accurate for stress episodes short relative to a monitoring cycle and pessimistic for a well-monitored book in calm markets, where it overstates the cushion required. The one-month horizon is a compromise and is varied in Section 6.2.

Remark 3 (Oracle prices). Aave liquidates against Chainlink prices that update on a heartbeat or deviation threshold. Discrete monitoring means a dip that recovers between updates does not liquidate, so the continuous barrier *overstates* liquidation frequency; but a stale oracle that gaps down liquidates at the gapped level, so the continuous model *understates* the recovery loss (compounding the jump overshoot of Section 7.3). We retain continuous monitoring for tractability. Calibration prices are reserve-level oracle-implied values (TVL divided by supplied balance), not the block-by-block feed.

Lemma 2.1 (Payoff factorization)**.**

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} \left(e^{h_0+Z_T} - b\right) \mathbf{1}_{\{\tau>T\}}, \tag{2}$$

and the factor $(e^{h_0+Z_T} - b)$ *is strictly positive on* $\{\tau > T\}$.

The payoff thus separates into a numeraire factor driven by S_2 and a barrier-killed claim on Z with moneyness set by h_0.

2.2. Optimization setup

Imposing the unit-initial-value constraint $n_1 S_{1,0} - n_2 S_{2,0} = 1$ and writing $x := n_2 S_{2,0}$ gives $e^{h_0} = b(1+x)/x$, hence the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \qquad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}, \qquad h_0 > \ln b, \tag{3}$$

and initial gross leverage

$$L_0 := n_1(h_0)\, S_{1,0} = \frac{e^{h_0}}{e^{h_0} - b}, \tag{4}$$

strictly decreasing from ∞ at $h_0 \downarrow \ln b$ to 1 (unlevered) as $h_0 \to \infty$. Holdings exist for $h_0 > \ln b$, but a position with $h_0 \le 0$ is liquidated at once and the protocol allows origination only at $h_0 \ge h_0^{\min} := \log(b/\mathrm{LTV}_{\max})$, so the feasible range is $h_0 \in [h_0^{\min}, \infty)$ ($h_0^{\min} = 0.066$ and $L_0 \le 3.70$ for WBTC collateral). Choosing h_0 is therefore equivalent to choosing leverage, and the portfolio problem reduces to $\max_{h_0 \ge h_0^{\min}} \Phi(h_0)$; we call the maximiser $h_0^\star$ the *optimal initial health buffer*.

The framework supports any criterion of the form

$$\Phi(h_0) \;=\; f\big(\mathbb{E}[W_T], \ldots, \mathbb{E}[W_T^K],\, p_{\mathrm{liq}}(h_0)\big)\,, \tag{5}$$

where W_T is terminal wealth (8), whose moments combine the raw killed moments $\mathbb{E}[\Pi_T^k]$ of Section 4 with the recovery of Section 7, $p_{\mathrm{liq}} := 1 - p_{\mathrm{surv}}$, and $K \max(\eta_{1,+}, \eta_{2,+}) < 1$ (Proposition 4.1). This covers mean–variance ($K = 2$), skewness/kurtosis penalties ($K = 4$), the Sharpe ratio and polynomial approximations to utility; unlike the mean–variance jump-diffusion criteria of Dang and Forsyth (2016) and Shen and Zou (2022), p_{liq} is a separately computed building block rather than an implicit risk charge.

We build the objective on *unconditional* wealth. The survival-conditional moments $\mathbb{E}[\Pi_T^k \mid \tau > T] = \mathbb{E}[\Pi_T^k]/p_{\mathrm{surv}}(h_0)$ are useful diagnostics but should not be maximised: conditioning keeps only favourable paths, so the conditional mean *rises* as the buffer thins toward the barrier (Table 3), rewarding leverage for a survivorship artefact. For illustration we use

$$\Phi(h_0) := \mathbb{E}[W_T] - \tfrac{\alpha}{2}\,\mathrm{Var}(W_T) - \delta\, p_{\mathrm{liq}}(h_0), \tag{6}$$

with risk aversion $\alpha \geq 0$ and an optional penalty $\delta \geq 0$. Since W_T already carries the liquidation loss, δ deliberately double-counts the event and so prices only its *non-pecuniary* costs (lost optionality, re-entry gas and slippage); $\delta = 0$ unless stated otherwise. Without recovery, $W_T = \Pi_T$.

2.3. The health-buffer evaluation map

The model's objective-independent output is, for every feasible h_0, the *evaluation map*

$$h_0 \longmapsto \left(p_{\text{liq}}(T,h_0),\, M_1(T,h_0),\ldots,M_K(T,h_0),\, \frac{M_1(T,h_0)}{p_{\text{surv}}(T,h_0)},\ldots,\frac{M_K(T,h_0)}{p_{\text{surv}}(T,h_0)} \right), \tag{7}$$

with $M_k(T,h_0) = \mathbb{E}[\Pi_T^k]$. No preference parameter enters it; a liquidation limit or a criterion such as (6) is applied afterwards. Section 7 extends the map with the moments of terminal wealth, the overshoot law and $\mathbb{E}[\log W_T]$. Adding a recovery value R_τ carried to T, investor wealth is

$$W_T \;=\; \Pi_T + R_\tau \mathbf{1}_{\{\tau \leq T\}}. \tag{8}$$

The separation matters in practice: the map is an exact function of its parameter inputs, whereas the selected buffer turns on the input the data estimate worst (Section 9).

3. Asset Dynamics

3.1. Bivariate Kou model

Under $\mathbb{P}$ each log-price satisfies

$$\mathrm{d}X_{i,t} = \mu_i^X \,\mathrm{d}t + \sigma_i \,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)} \right), \qquad i = 1,2, \tag{9}$$

where $\mathrm{d}\langle B_1, B_2 \rangle_t = \rho \,\mathrm{d}t$, N_i is Poisson with intensity λ_i independent of (B_1,B_2), and the jumps $J_i^{(n)}$ are i.i.d. double-exponential with density, characteristic function and moment generating function

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \qquad \varphi_{J_i}(u) = \frac{p_i}{1-iu\eta_{i,+}} + \frac{1-p_i}{1+iu\eta_{i,-}}, \tag{10}$$

$$M_i(s) = \frac{p_i}{1-s\eta_{i,+}} + \frac{1-p_i}{1+s\eta_{i,-}}, \qquad s \in \left(-\tfrac{1}{\eta_{i,-}}, \tfrac{1}{\eta_{i,+}} \right). \tag{11}$$

Here $\eta_{i,\pm}$ are the *mean* jump sizes (not rates), with $\eta_{i,+} < 1$. We parametrise by the continuously compounded price growth rate μ_i, $\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. With the price-jump compensator

$$\chi_i := M_i(1) - 1 = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1, \tag{12}$$

the log-price drift is

$$\mu_i^X := \mu_i - \tfrac{1}{2}\sigma_i^2 - \lambda_i \chi_i. \tag{13}$$

(X_1,X_2) is a Lévy process with $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t}+vX_{2,t})}] = e^{t\Psi(u,v)}$, where

$$\begin{aligned} \Psi(u,v) := {} & i(\mu_1^X u + \mu_2^X v) - \tfrac{1}{2}\left(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv \right) \\ & + \lambda_1 \left(\varphi_{J_1}(u) - 1 \right) + \lambda_2 \left(\varphi_{J_2}(v) - 1 \right). \end{aligned} \tag{14}$$

Table 1 collects the notation.

Symbol	Meaning	Constraint
μ_i	Continuously compounded expected price growth: $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$	
σ_i, ρ	Diffusion volatility of X_i; Brownian correlation	$\sigma_i > 0$, $\lvert\rho\rvert < 1$
λ_i, p_i	Jump intensity; probability a jump is upward	$\lambda_i > 0$, $p_i \in (0,1)$
$\eta_{i,\pm}$	Mean upward / downward jump size	$0 < \eta_{i,+} < 1$, $\eta_{i,-} > 0$
χ_i, μ_i^X	Jump compensator (12); log-price drift (13)	
b	Liquidation threshold (max. borrow-to-collateral ratio before liquidation)	$b \in (0,1)$
T, $S_{i,0}$, n_i	Horizon; initial prices; coin holdings	positive
h_0	Initial log health factor; the sizing variable, determines n_i via (3)	$h_0 > \log b$
τ, Π_T	Liquidation time; surviving-path terminal wealth	

Table 1: Model inputs (top) and portfolio quantities (bottom).

3.2. Tilted dynamics

For each integer $k \geq 0$ with $k\eta_{2,+} < 1$ define the k-tilted measure $\mathbb{P}^{(k)}$ by

$$\left.\frac{\mathrm{d}\mathbb{P}^{(k)}}{\mathrm{d}\mathbb{P}}\right|_{\mathscr{F}_t} = \exp\left(kX_{2,t} - t\,\Psi(0,-ik)\right).$$

$k = 0$ recovers $\mathbb{P}$; a superscript (k) always denotes evaluation under $\mathbb{P}^{(k)}$.

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$)**.** *Under $\mathbb{P}^{(k)}$, $Z = X_1 - X_2$ is a Lévy process with Laplace exponent*

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \tfrac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big), \tag{15}$$

where $\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ and $\mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$. Its generator satisfies

$$\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s)\, e^{sz}. \tag{16}$$

▷ *Proof, p. 28*

Under the tilt the jumps of Z have phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, upward jumps of X_2 are downward jumps of Z, and the tilt moves their rate to $1/\eta_{2,+} - k$, positive precisely when $k\eta_{2,+} < 1$.

Definition 3.1 (Phase operators)**.** For $j = 1, 2$ let

$$K_{j,+}^{(k)}[f](z) := \int_0^\infty f(z+y)\, r_{j,+}^{(k)} e^{-r_{j,+}^{(k)} y}\,\mathrm{d}y, \qquad K_{j,-}^{(k)}[f](z) := \int_{-\infty}^0 f(z+y)\, r_{j,-}^{(k)} e^{r_{j,-}^{(k)} y}\,\mathrm{d}y.$$

For $-r_{j,-}^{(k)} < \gamma < r_{j,+}^{(k)}$ they act on exponentials as

$$K_{j,+}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma}\, e^{\gamma z}, \qquad K_{j,-}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma}\, e^{\gamma z}. \tag{17}$$

This construction is due to Kou and Wang (2003) and is the identity that Sepp (2004) uses to reduce barrier pricing to a finite linear system; Section 5 uses it the same way. On a half-line the downward operator acts on the zero-extended solution, where the integral is over a finite range and (17) holds up to a boundary term for every $\gamma \neq -r_{j,-}^{(k)}$ (proof of Proposition 5.2).

4. Moments via Measure Change

Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly couples (X_1, X_2) to the barrier event. Tilting by $e^{kX_{2,T}}$ absorbs the short-leg factor into the density and leaves a one-dimensional killed expectation of Z, which remains Lévy under $\mathbb{P}^{(k)}$. Let

$$g_k(z;h_0) := (e^{h_0+z} - b)^k,$$

positive on $\{\tau > T\}$ and a polynomial in e^{Z_T}. By (2), $\Pi_T^k = (n_2 S_{2,0}/b)^k e^{kX_{2,T}} g_k(Z_T;h_0)\mathbf{1}_{\{\tau>T\}}$, and changing measure gives

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2 S_{2,0}}{b}\right)^k e^{T\Psi(0,-ik)}\, \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T;h_0)\,\mathbf{1}_{\{\tau>T\}}\Big]. \tag{18}$$

Proposition 4.1 (Every admissible integer-order moment)**.** *For every integer $k \in \mathscr{K} := \{n \in \mathbb{N} : n\max(\eta_{1,+}, \eta_{2,+}) < 1\}$, with $m_k(T;h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}[g_k(Z_T;h_0)\mathbf{1}_{\{\tau>T\}}]$ and (3),*

$$\mathbb{E}[\Pi_T^k] = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0}-b)^k}\, m_k(T;h_0),$$

where $m_k(T;h_0)$ is computed by the Laplace resolvent of Section 5.

The two conditions in $\mathscr{K}$ play different roles. $k\eta_{1,+} < 1$ is needed for the moment to be finite at all, since $\Pi_T \geq (1-b)n_1 S_{1,T}$ on survival, and for $\psi_Z^{(k)}(j)$, $j \leq k$, to be finite in the particular solution below. $k\eta_{2,+} < 1$ is needed by the tilt. At the headline calibration the binding condition is the second ($k \leq 88$).

With $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$, the objective (6) without recovery is $\mathbb{E}[\Pi_T] - \frac{\alpha}{2}(\mathbb{E}[\Pi_T^2] - \mathbb{E}[\Pi_T]^2) - \delta\, p_{\text{liq}}(h_0)$; with recovery, $\mathbb{E}[\Pi_T^k]$ is replaced by the wealth moments of Section 7.

Remark 4 (No-shorting limit). As $h_0 \to \infty$, $n_2 S_{2,0} \to 0$ and $n_1 S_{1,0} \to 1$, so liquidation becomes impossible and

$$\lim_{h_0\to\infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik,0)},$$

which equals $e^{\mu_1 T}$ for $k = 1$ by (13). The formula must reproduce this limit exactly, which provides a numerical check.

5. Resolvent Equations for Moments and Survival Probability

Proofs of the results in Sections 3.2–5 are in Appendix D.

5.1. Killed moment resolvent

Fix $k \in \mathscr{K}$. The killed moment function $U(t,z) := \mathbb{E}_z^{(k)}[g_k(Z_t;h_0)\mathbf{1}_{\{\tau>t\}}]$, $z > -h_0$, with $U = 0$ for $z \leq -h_0$, solves $\partial_t U = \mathscr{L}^{(k)} U$ with $U(0,z) = g_k(z;h_0)$. Its Laplace transform $\widehat{U}(q,z) := \int_0^\infty e^{-qt}U(t,z)\,\mathrm{d}t$ therefore satisfies, after integrating by parts in t, the resolvent equation

$$(q - \mathscr{L}^{(k)})\widehat{U}(q,z) = g_k(z;h_0), \qquad z > -h_0, \tag{19}$$

with $\widehat{U}(q,z) = 0$ for $z \leq -h_0$.

Assumption 5.1 (Generic six-root regime)**.** Fix $k \in \mathscr{K} \cup \{0\}$ and $q > 0$. Assume:

(i) $\sigma_Z > 0$;

(ii) $\psi_Z^{(k)}(\gamma) = q$ has six simple roots with $\mathrm{Re}(\gamma_{1,2,3}) > 0$ and $\mathrm{Re}(\gamma_{4,5,6}) < 0$;

(iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \dots, k$.

For real $q > 0$ with distinct poles ($r_{1,+}^{(k)} \neq r_{2,+}^{(k)}$ and $r_{1,-}^{(k)} \neq r_{2,-}^{(k)}$) the three-and-three split is automatic. Each Kou term of $\psi_Z^{(k)}$ changes sign through its pole, tending to $+\infty$ on the side facing the origin and to $-\infty$ on the other; $\psi_Z^{(k)}(0) - q = -q < 0$; and the diffusion term sends $\psi_Z^{(k)} - q$ to $+\infty$ as $|\gamma| \to \infty$. By the intermediate value theorem there is a real root in each of the six intervals cut out by the origin and the poles $\{r_{1,+}^{(k)}, r_{2,+}^{(k)}, -r_{1,-}^{(k)}, -r_{2,-}^{(k)}\}$, three on each side, and since clearing denominators gives a polynomial of degree six these are all the roots and they are simple (Kou and Wang, 2003; Sepp, 2004; Cai and Kou, 2011). The assumption has content for the complex q at which the Talbot contour evaluates the transform, part of which lies in $\mathrm{Re}\, q < 0$. There the roots are continued analytically from the real case, and the implementation classifies them by the rank of their real parts rather than by sign. At the headline calibration the sign split holds at every Talbot node, and node counts from $M = 16$ to 48 agree to 10^{-8}.

On $z > -h_0$ the forcing expands as

$$g_k(z; h_0) = \sum_{j=0}^{k} c_{k,j}(h_0)\, e^{jz}, \qquad c_{k,j}(h_0) := \binom{k}{j} (-1)^{k-j} b^{k-j} e^{jh_0}. \tag{20}$$

Lemma 5.1 (Particular and homogeneous solutions)**.** *Under Assumption 5.1,*

$$\widehat{U}_{\mathrm{part}}(q, z; h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)}\, e^{jz} \tag{21}$$

solves (19), and the exponential solutions of $(q - \mathscr{L}^{(k)})u = 0$ on $z > -h_0$ that are bounded at $+\infty$ are

$$\widehat{U}_{\mathrm{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0)\, e^{\gamma_{m+3}(q)(z+h_0)}. \tag{22}$$

▷ *Proof, p. 28*

Proposition 5.2 (Barrier resolvent)**.** *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \tag{23}$$

$\widehat{U} = \widehat{U}_{\mathrm{part}} + \widehat{U}_{\mathrm{hom}}$ with $\mathbf{B}^{(k)} := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ the unique solution of $\mathbf{M}_{\mathrm{bar}}^{(k)}(q)\, \mathbf{B}^{(k)} = -\mathbf{b}^{(k)}(q; h_0)$, where

$$\mathbf{M}_{\mathrm{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6} \\ \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6} \end{pmatrix}, \qquad \mathbf{b}^{(k)}(q; h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)\, e^{-jh_0}}{q - \psi_Z^{(k)}(j)} \begin{pmatrix} 1 \\ \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + j} \\ \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + j} \end{pmatrix}.$$

The rows of $\mathbf{b}^{(k)}$ are $\widehat{U}_{\mathrm{part}}$ and $K_{j,-}^{(k)}[\widehat{U}_{\mathrm{part}}]$, $j = 1, 2$, evaluated at $z = -h_0$. The killed moment is $m_k(T; h_0) = \mathscr{L}_{q \to T}^{-1}[\widehat{U}(q, 0; h_0)]$ with

$$\widehat{U}(q, 0; h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} + \sum_{m=1}^{3} B_m^{(k)}(q; h_0)\, e^{\gamma_{m+3}(q) h_0}. \tag{24}$$

▷ *Proof, p. 28*

Lemma 5.3 (Invertibility)**.** *Under Assumption 5.1 and (23), $\mathbf{M}^{(k)}_{\mathrm{bar}}(q)$ is invertible for every admissible $k \geq 0$.*

▷ *Proof, p. 28*

5.2. Survival probability

The survival probability is the $k = 0$, $g \equiv 1$ case of the same construction. Its transform $\widehat{U}_{\mathrm{surv}}(q, z; h_0) := \int_0^\infty e^{-qt}\mathbb{P}_z(\tau > t)\,\mathrm{d}t$ solves $(q - \mathscr{L}^{(0)})\widehat{U}_{\mathrm{surv}} = 1$ on $z > -h_0$, vanishing below.

Proposition 5.4 (Survival resolvent)**.** *Under Assumption 5.1 with $k = 0$ and*

$$r^{(0)}_{1,-} \neq r^{(0)}_{2,-}, \quad \textit{i.e.} \quad \eta_{1,-} \neq \eta_{2,+}, \tag{25}$$

let $\gamma^{(0)}_{4,5,6}(q)$ be the roots of $\psi^{(0)}_Z(\gamma) = q$ with negative real part. Then

$$\widehat{U}_{\mathrm{surv}}(q, z; h_0) = \frac{1}{q} + \sum_{m=1}^{3} C_m(q)\, e^{\gamma^{(0)}_{m+3}(q)(z+h_0)}, \qquad \mathbf{M}^{(0)}_{\mathrm{bar}}(q)\,\mathbf{C}(q) = -\tfrac{1}{q}(1,1,1)^\top,$$

and $p_{\mathrm{surv}}(T; h_0) = \mathscr{L}^{-1}_{q\to T}[\widehat{U}_{\mathrm{surv}}(q, 0; h_0)]$, $p_{\mathrm{liq}} = 1 - p_{\mathrm{surv}}$.

▷ *Proof, p. 29*

6. Numerical Results

We validate the resolvent against Monte Carlo (MC) for $k = 0, \ldots, 4$ — an illustration, not a limit: the same code handles every $k \in \mathscr{K}$ — and check the no-shorting limit. Inversion uses the Talbot contour (Talbot, 1979) with $M = 32$ nodes.

Parameters (Table 2) are calibrated by ECF matching (Appendix A) to 840 four-hour WBTC (asset 1) and WETH (asset 2) log-returns on Aave v3 Ethereum after subtracting an exponentially weighted mean, with drifts adjusted for Aave supply/borrow APRs. The liquidation threshold $b = 0.78$ and maximum origination LTV 0.73 give a minimum feasible buffer $h_0^{\min} = \log(0.78/0.73) \approx 0.066$. WBTC and WETH have distinct downward phase rates, so (23) holds for every admissible k. The ECF is fitted to de-meaned returns, so the fitted μ_i are price-growth normalizers, not estimates of the window's drift: their difference $\mu_1 - \mu_2 = -0.86$ even has the opposite sign to the window's realised spread drift, which is $+0.13$ per year in price-growth terms ($+0.10$ net of carry) and $+0.19$ in raw log terms (from the two raw log drifts in Table 2). The tables of this section use the calibrated normalizers as stated; Section 9 uses sample drifts.

For $h_0 \in \{0.07, 0.10, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00\}$ Table 3 reports p_{surv} and the survival-conditional mean, variance, skewness and excess kurtosis of Π_T, computed from $\mu_k := \mathbb{E}[\Pi_T^k \mid \tau > T]$ in the usual way, together with the $h_0 \to \infty$ limits $\mu_k \to \exp(T\Psi(-ik, 0))$ of Remark 4.

The conditional moments agree with MC to within 0.005 (mean), 0.0003 (variance), 0.016 (skewness) and 0.07 (excess kurtosis) at every tabulated buffer. In this run the Laplace method took about 0.14 s for all five inversions against 61–66 s for MC, a speed-up of about 440×; timings are machine- and load-dependent, so only the order of magnitude is meaningful. The only systematic discrepancy is in survival: MC exceeds Laplace by about one point near $h_0^{\min}$ and by nothing for $h_0 \geq 0.5$. Section 6.1 shows this is MC discretisation, not analytic error.

Parameter	WBTC	WETH
ECF price-growth normalizer μ_i	0.302	1.161
Log-price drift μ_i^X	-0.575	0.222
Model log-return drift $\mathbb{E}[\mathrm{d}X_i]/\mathrm{d}t$	0.262	1.072
Empirical de-meaned log drift	0.234	0.297
Realised (raw) log drift	-0.390	-0.578
Diffusion volatility σ_i	0.264	0.328
Jump intensity λ_i	358.29	450.65
Upward-jump probability p_i	0.770	0.457
Mean upward / downward jump $\eta_{i,+}$ / $\eta_{i,-}$	0.00413 / 0.00367	0.01126 / 0.00601
Initial price $S_{i,0}$	64,125.20	1,872.81
Brownian correlation ρ; threshold b; horizon T	0.847; 0.78; $1/12$	

Table 2: Calibrated parameters: 840 four-hour WBTC/WETH log-returns, Aave v3 Ethereum, 2026-03-17 to 2026-08-04, EWM de-meaning (span 30). Prices are oracle-implied USD values (reserve TVL / supplied balance) resampled to four hours; initial prices are the last sample (2026-08-04 16:00 UTC). Drifts include realised average APRs: $\mu_1 \mathrel{+}= 0.0061\%$ (WBTC supply), $\mu_2 \mathrel{+}= 2.5104\%$ (WETH borrow). The empirical de-meaned price-growth rates are 0.303 and 0.421, so the fitted WETH μ_2 is a weakly identified normalizer, not a return forecast.

6.1. An accuracy-matched Monte Carlo benchmark

The Euler–Poisson scheme checks the barrier only at step ends, so it misses intra-step crossings and *overstates* survival. An accuracy-matched simulator draws the diffusion on a grid and adds the Brownian-bridge probability of an intra-step crossing analytically (Glasserman, 2004), with jumps applied and checked at step ends. At $h_0 = 0.07$ (Table 4) the bridge-corrected estimate is within two standard errors of the Laplace value 0.33029 at every tabulated step count, and within 0.2 standard errors at all but one, while the naive check is biased by $+0.011$ at 125 steps and still $+0.003$ at 4,000. The analytic survival probability is therefore unbiased. Talbot inversion is also stable: varying M from 16 to 48 changes $p_{\mathrm{surv}}(h_0{=}0.07)$ by less than 10^{-8}. The bridge-corrected simulator is already unbiased at 125 steps, where 200,000 paths take 5.6–8.6 s across our runs for a standard error of 10^{-3}, against 16–20 ms for the noise-free Laplace survival probability: again two to three orders of magnitude, and without sampling error.

6.2. Parameter sensitivity

Figure 1 sweeps each parameter in turn. At $H_0 = 1.2$ (benchmark survival 0.83) the *horizon* matters most: survival is 1.00 at one week and 0.32 at three months. *Jump intensity* and *diffusion volatility* come next: scaling the intensities from 0.1 to 2 times the benchmark moves survival from 0.96 to 0.73, and scaling σ_1, σ_2 from 0.3 to 2 times moves it from 0.88 to 0.71. Jumps carry about three quarters of the spread variance at the benchmark, which is why the intensity sweep matters as much as the volatility sweep. *Correlation* acts only through the spread variance: lowering ρ from 0.85 to 0.5 cuts survival to 0.74, and the conditional mean barely moves. Jump intensity and horizon also raise the conditional mean at low H_0 through survivor selection. Survival does not depend on b at fixed H_0 — only the payoff kernel does — so row (e) plots leverage $L_0 = H_0/(H_0 - b)$, which reaches $\approx 6.3\times$ at $H_0^{\min}$ for $b = 0.90$.

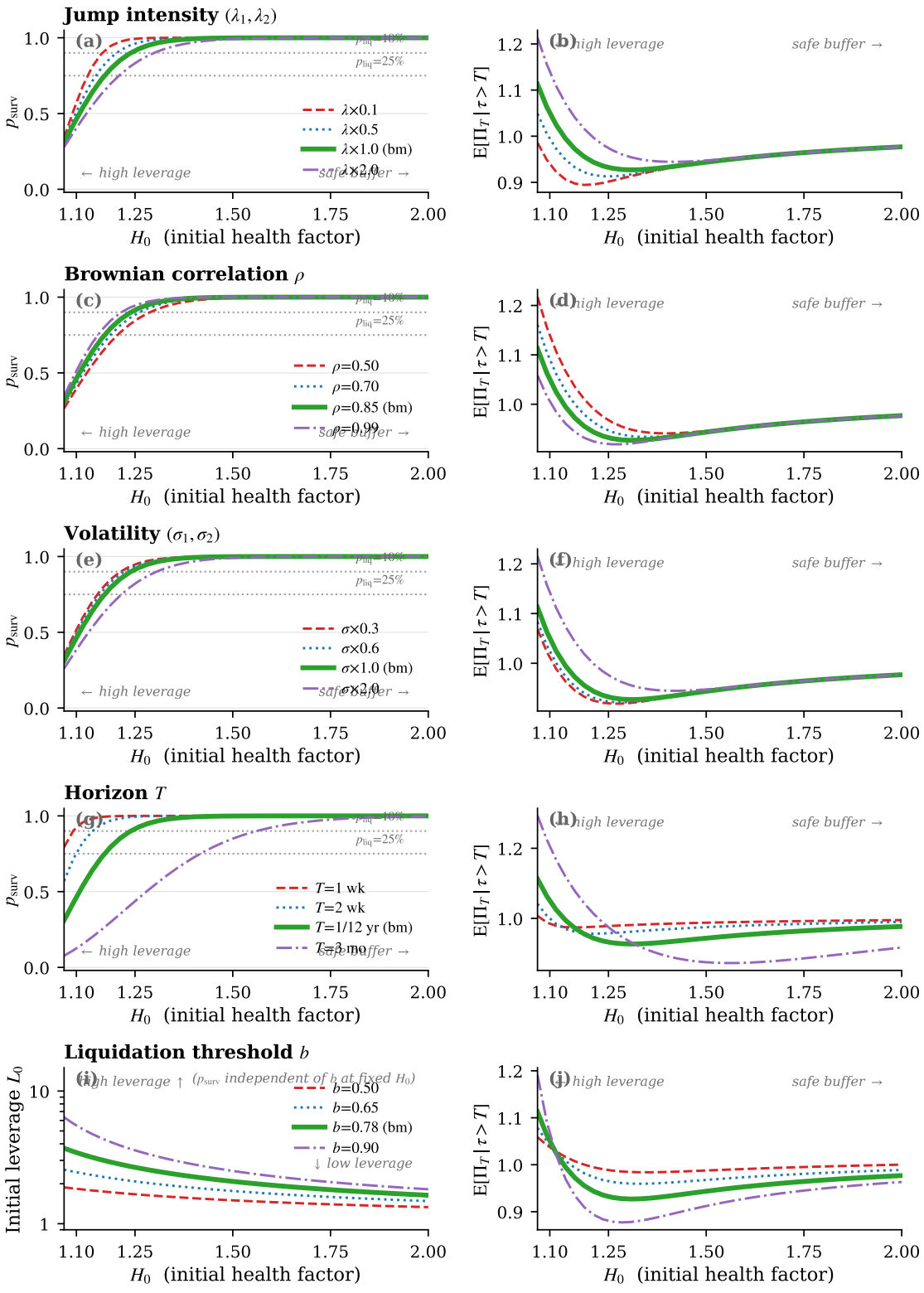


Figure 1: Left: survival probability (rows a–d) and initial leverage L_0 (row e, log scale) versus H_0. Right: conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$. Each row varies one parameter, others at Table 2; the heavier solid line is the benchmark. $H_0 \in [1.068, 2.000]$, 40 points per curve.

		p_{surv}					
h_0	H_0	Lap.	MC	Mean	Variance	Skewness	Ex. Kurt.
0.07	1.07	0.3303	0.3421	1.1035	0.0322	0.6535	0.4790
0.10	1.11	0.4879	0.4985	1.0413	0.0306	0.6320	0.3989
0.25	1.28	0.9505	0.9524	0.9278	0.0287	0.2692	−0.1028
0.50	1.65	1.0000	0.9999	0.9573	0.0169	0.0955	0.1592
0.75	2.12	1.0000	1.0000	0.9812	0.0114	0.2026	0.1463
1.00	2.72	1.0000	1.0000	0.9949	0.0092	0.2514	0.1444
1.50	4.48	1.0000	1.0000	1.0095	0.0078	0.2724	0.1428
2.00	7.39	1.0000	1.0000	1.0166	0.0074	0.2705	0.1380
∞	∞	1.0000	—	1.0255	0.0072	0.2627	0.1286

Table 3: Laplace–resolvent survival and survival-conditional statistics for the calibrated WBTC/WETH book. MC uses 10^5 Euler–Poisson paths with 252 steps over the month (seed 42). The bottom row is the analytical no-shorting limit; Laplace at $h_0 = 100$ matches it to relative error below 3×10^{-6}.

Steps	bridge-corrected p_{surv}	naive grid p_{surv}	naive bias
125	0.3303 (z=−0.01)	0.3408	+0.0106
500	0.3304 (z=−0.13)	0.3367	+0.0064
1,000	0.3324 (z=−1.99)	0.3344	+0.0041
4,000	0.3302 (z=+0.11)	0.3333	+0.0030

Table 4: Survival at $h_0 = 0.07$ against steps over the month; 200,000 paths, MC standard error $\approx 1.1 \times 10^{-3}$, Laplace reference 0.33029; z is the standardised gap to Laplace. Script `mc_convergence.py`.

7. Liquidation Recovery, Overshoot and Log Wealth

This section supplies R_τ in (8) under the protocol's liquidation convention, at the cost of one additional Laplace inversion per moment order and with the barrier system unchanged. The same objects then give the law of the overshoot, the probability of bad debt and the expected log of terminal wealth.

7.1. Recovery at the barrier

A liquidator repays debt and seizes collateral worth $(1+\ell)$ times the repayment, ℓ being the liquidation bonus (Aave v3 Ethereum: 0.05 for WETH, 0.045 for major stablecoins, 0.05–0.075 for wrapped BTC). Full repayment leaves $R_\tau = V_{1,\tau} - (1+\ell)V_{2,\tau}$. On a continuous crossing $V_{2,\tau} = bV_{1,\tau}$, so

$$R_\tau \;=\; V_{2,\tau}\,\frac{1-b(1+\ell)}{b}, \tag{26}$$

positive when $\ell < 1/b - 1$ (0.282 at $b = 0.78$). Writing recovery through the *debt* leg makes it proportional to $e^{X_{2,\tau}}$, the factor the framework already tilts by.

Remark 5 (Continuous-crossing approximation). A downward jump can carry H strictly below one, leaving less recovery than (26), possibly negative (bad debt). Equation (26) is therefore an upper bound; Section 7.3 computes the exact value.

7.2. *The first-passage transform*

We need $\mathbb{E}[e^{kX_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}]$ for $k=1,2$. Since $\exp(kX_{2,t}-t\Psi_k)$, $\Psi_k:=\Psi(0,-ik)$, is a martingale, changing measure on $\mathscr{F}_\tau$ gives $\mathbb{E}[e^{kX_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}]=\mathbb{E}^{(k)}[e^{\Psi_k\tau}\mathbf{1}_{\{\tau\le T\}}]$, with Laplace transform

$$\int_0^\infty e^{-qT}\,\mathbb{E}\Big[e^{kX_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}\Big]\,\mathrm{d}T=\frac{1}{q}\Big(1-(q-\Psi_k)\,\widehat{p}^{(k)}_{\mathrm{surv}}(q-\Psi_k)\Big),\qquad q>\Psi_k, \tag{27}$$

where $\widehat{p}^{(k)}_{\mathrm{surv}}$ is the survival transform of Proposition 5.4 under the k-tilted dynamics. Because Π_T and $R_\tau\mathbf{1}_{\{\tau\le T\}}$ live on disjoint events,

$$\mathbb{E}[W_T]=\mathbb{E}[\Pi_T]+\mathbb{E}[R_\tau\mathbf{1}_{\{\tau\le T\}}],\qquad \mathbb{E}[W_T^2]=\mathbb{E}[\Pi_T^2]+\mathbb{E}[R_\tau^2\mathbf{1}_{\{\tau\le T\}}]. \tag{28}$$

At $k=0$, (27) returns p_{liq} and agrees with Proposition 5.4 to 1.7×10^{-11}.

Recovery is material. At $H_0=1.16$ it raises mean terminal wealth from 0.694 to 0.856 and cuts its variance from 0.217 to 0.057 by removing the total-loss atom at zero. It does not change the sizing conclusion of Section 9.

7.3. *Exact recovery under jump overshoot*

With the actual leg values at liquidation and $h_\tau:=h_0+Z_\tau\le 0$ (zero at a continuous crossing),

$$R_\tau\;=\;V_{2,\tau}\left(\frac{e^{h_\tau}}{b}-(1+\ell)\right), \tag{29}$$

which is smaller than (26) whenever $h_\tau<0$. Rather than integrating against the (exponential-mixture) overshoot density, we note that per unit initial equity

$$R_\tau=A\,e^{X_{1,\tau}}-B\,e^{X_{2,\tau}},\qquad \mathrm{Hit}(a_1,a_2):=\mathbb{E}\left[e^{a_1X_{1,\tau}+a_2X_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}\right], \tag{30}$$

with $A=H_0/(H_0-b)$ and $B=(1+\ell)b/(H_0-b)$, so $\mathbb{E}[R_\tau\mathbf{1}]=A\,\mathrm{Hit}(1,0)-B\,\mathrm{Hit}(0,1)$ and $\mathbb{E}[R_\tau^2\mathbf{1}]=A^2\mathrm{Hit}(2,0)-2AB\,\mathrm{Hit}(1,1)+B^2\mathrm{Hit}(0,2)$. The tilted first-passage law already integrates over the landing point, so overshoot is captured exactly. Tilting by $\exp(a_1X_{1,t}+a_2X_{2,t}-t\Psi_{\mathrm{M}}(a_1,a_2))$, with $\Psi_{\mathrm{M}}(a_1,a_2):=\log\mathbb{E}[e^{a_1X_{1,1}+a_2X_{2,1}}]$, gives

$$\int_0^\infty e^{-qT}\,\mathrm{Hit}(a_1,a_2)\,\mathrm{d}T=\frac{1}{q}\Big(1-(q-\Psi_{\mathrm{M}})\,\widehat{p}^{(a_1,a_2)}_{\mathrm{surv}}(q-\Psi_{\mathrm{M}})\Big),\qquad q>\Psi_{\mathrm{M}}(a_1,a_2), \tag{31}$$

where $\widehat{p}^{(a_1,a_2)}_{\mathrm{surv}}$ is the survival transform under the (a_1,a_2)-tilted dynamics of Z, with exponent $\psi_Z^{(a_1,a_2)}(s)=\Psi_{\mathrm{M}}(a_1+s,a_2-s)-\Psi_{\mathrm{M}}(a_1,a_2)$ and phase rates $r_{1,+}=1/\eta_{1,+}-a_1$, $r_{1,-}=1/\eta_{1,-}+a_1$, $r_{2,+}=1/\eta_{2,-}+a_2$, $r_{2,-}=1/\eta_{2,+}-a_2$; $(a_1,a_2)=(0,k)$ recovers Lemma 3.1. The barrier system stays 3×3. $\mathrm{Hit}(0,k)$ reproduces (27) to 6×10^{-14} and $\mathrm{Hit}(0,0)$ reproduces $1-p_{\mathrm{surv}}$ to 3×10^{-11}; against an MC recording leg values at the crossing step, all five moments agree up to a discretisation bias that shrinks with the step count.

At the headline calibration's jump sizes ($\eta\approx0.4$–1.1%) the correction is small: at $H_0=1.16$ mean recovery falls from 0.162 to 0.157 (-3.1%) and wealth variance rises from 0.057 to 0.060. To see when overshoot bites, Table 5 scales the jumps two ways. Design A scales the mean jump size by s at fixed intensity, which multiplies jump variance by s^2 — a far more volatile market. The correction reaches -58% at $s=10$ and mean recovery turns negative at $s=20$: the continuous formula reports positive recovery where the protocol absorbs bad debt. Design B also divides intensity by s^2, holding

jump variance fixed — genuinely rarer, larger jumps. The correction is milder (-36% at $s = 20$) and mean recovery stays positive. The driver is jump size rather than jump variance: at fixed variance the correction still reaches -36%, while the headline jumps carry about three quarters of the spread variance yet give only -3% because each is small. Throughout, R_τ is the signed residual equity; a non-recourse borrower receives R_τ^+, which differs from it only on bad-debt paths, whose probability Section 7.4 gives in closed form. The headline calibration produces its kurtosis from many small jumps (cf. the identification ridge of Ramezani and Zeng 2007 and Appendix A.2), so there the correction is small. It is not small under every calibration: the full-history daily fit of Appendix A.3, whose jumps are fewer and larger ($\eta_{2,+} \approx 3.2\%$), gives a correction of -11% at $H_0 = 1.16$.

	Design A: variance $\propto s^2$		Design B: variance fixed	
Jump scale s	$\mathbb{E}[R_\tau]$ exact	rel. corr.	$\mathbb{E}[R_\tau]$ exact	rel. corr.
$1\times$ (fitted)	0.157	-3.1%	0.157	-3.1%
$3\times$	0.263	-15.1%	0.129	-9%
$10\times$	0.175	-58.4%	0.065	-26%
$20\times$	-0.090	-119.6%	0.041	-36%

Table 5: Exact mean recovery at $H_0 = 1.16$ and its correction relative to the continuous-crossing value (26). The continuous-crossing *expected* recovery still moves with p_{liq}: under Design A p_{liq} rises from 0.29 to 0.70 (continuous mean recovery 0.162 $\to$ 0.461); under Design B it falls to 0.11 ($\to$ 0.064). Script `overshoot_table.py`.

7.4. The overshoot law and the probability of bad debt

The joint hitting moments integrate over the landing point without exposing it. The landing point itself has a simple law. Write $u := -h_\tau \geq 0$ for the overshoot of the log health factor below the barrier.

Proposition 7.1 (Overshoot law)**.** *For every T there are weights $P_0, P_1, P_2 \geq 0$, with $P_0 + P_1 + P_2 = p_{\text{liq}}(T)$, such that for every bounded measurable f*

$$\mathbb{E}\big[f(u)\,\mathbf{1}_{\{\tau \leq T\}}\big] = P_0\, f(0) + \sum_{i=1}^{2} P_i \int_0^\infty f(u)\, r_i e^{-r_i u}\, \mathrm{d}u, \qquad r_1 = \frac{1}{\eta_{1,-}}, \quad r_2 = \frac{1}{\eta_{2,+}}. \tag{32}$$

The weights solve the 3×3 system $e^{jh_0}\mathrm{Hit}(j,-j) = P_0 + P_1\, r_1/(r_1 + j) + P_2\, r_2/(r_2 + j)$ for $j = 0, 1, 2$.

▷ *Proof, p. 29*

Z reaches the barrier either continuously (the diffusion creeps across, $u = 0$) or by a downward jump, which is a down-jump of X_1 (rate r_1) or an up-jump of X_2 (rate r_2). By memorylessness, the excess of a crossing jump over the distance to the barrier is exponential with the rate of that jump, whatever the path before it (Kou, 2002; Kou and Wang, 2003); Appendix D makes this precise with the compensation formula. This gives (32). Taking $f(u) = e^{-ju}$, with $e^{-ju} = e^{jh_0} e^{jZ_\tau}$ and Hit from (31), yields the system. At the benchmark the weights reproduce the hitting moments at $j = 5$ and $j = 10$ to about 10^{-10}, which checks the structure independently.

Recovery is non-positive exactly when $u \geq u^\star := -\log\big(b(1+\ell)\big)$ (0.200 at $b = 0.78$, $\ell = 0.05$). On those paths the protocol absorbs bad debt and a non-recourse borrower ends with nothing. Proposition 7.1 gives the bad-debt probability in closed form,

$$\mathbb{P}(R_\tau \leq 0,\ \tau \leq T) = P_1 e^{-r_1 u^\star} + P_2 e^{-r_2 u^\star}. \tag{33}$$

It is 3×10^{-9} at $H_0 = 1.16$ under the headline calibration, where jumps are below one percent, and 3×10^{-4} under the full-history daily calibration.

7.5. *Expected log wealth*

Moments of every admissible order cover polynomial criteria. The log of wealth, which defines the growth-optimal (Kelly) investor (Kelly, 1956; MacLean et al., 2011), is not polynomial, but it is also within reach. Per unit equity, $\Pi_T = e^{X_{2,T}}(e^{h_T} - b)/(H_0 - b)$ on survival and $R_\tau = e^{X_{2,\tau}}(e^{-u} - c)/(H_0 - b)$ on liquidation, with $c := b(1+\ell)$. Since $\log(e^h - b) = h - \sum_{n\geq 1}(b^n/n)e^{-nh}$ for $h \geq 0$,

$$\begin{aligned}\mathbb{E}[\log W_T] = \mathbb{E}\big[(X_{2,T} + h_T)\mathbf{1}_{\{\tau>T\}}\big] - \sum_{n\geq 1}\frac{b^n}{n}\,e^{-nh_0}S(-n,n) - \log(H_0 - b)\\ + \mathbb{E}\big[X_{2,\tau}\mathbf{1}_{\{\tau\leq T\}}\big] + \mathbb{E}\big[\log(e^{-u} - c)\,\mathbf{1}_{\{\tau\leq T\}}\big],\end{aligned} \tag{34}$$

where $S(a_1,a_2) := \mathbb{E}[e^{a_1X_{1,T}+a_2X_{2,T}}\mathbf{1}_{\{\tau>T\}}]$ is a killed joint moment. Its Laplace transform is the tilted survival transform at $q - \Psi_{\mathrm{M}}(a_1,a_2)$, by the change of measure behind (31) applied on $\mathscr{F}_T$. The linear terms are tilt derivatives of S and Hit at zero, and the last term is a one-dimensional integral against (32). The series converges geometrically because $be^{-h_T} \leq b < 1$ on survival. The tilt $(-n,n)$ is admissible only for $n < \min(1/\eta_{2,+}, 1/\eta_{1,-})$, which caps the series under heavy jump tails. Since $\mathbb{E}[e^{-nh_T}\mathbf{1}_{\{\tau>T\}}]$ decreases in n, the truncation error after N terms is at most $\mathbb{E}[e^{-Nh_T}\mathbf{1}_{\{\tau>T\}}]\,b^{N+1}/((N+1)(1-b))$, which is below 10^{-5} in every case reported. The last term of (34) is taken over paths without bad debt; the next paragraph treats the rest.

Two features matter. First, under the zero-payoff convention $\log W_T = -\infty$ on every liquidated path, so a Kelly investor never levers; recovery is what makes the criterion usable. Second, on bad-debt paths $W_T = 0$ even with recovery, so strict Kelly is $-\infty$ at every levered buffer whenever (33) is positive. On those paths we replace $\log(e^{-u} - c)$ in (34) by $\log 10^{-4}$, which floors wealth at $10^{-4}e^{X_{2,\tau}}/(H_0 - b)$, and report (33) alongside. Kelly quantities below refer to this floored objective. The floor is immaterial at the headline calibration and of order 10^{-3} in $\mathbb{E}[\log W_T]$ at the thinnest buffer under the full-history one. Evaluating (34) takes about 0.3 s per buffer. A bridge-corrected Monte Carlo that records the exact leg values at liquidation agrees to within 1.6 standard errors at $H_0 \in \{1.07, 1.16, 1.30\}$, and as $h_0 \to \infty$ the expression tends to $\mathbb{E}[X_{1,T}]$ to 10^{-6}.

8. Carry-Adjusted Position Sizing

On Aave the collateral earns a supply rate and the debt accrues a borrow rate, both compounding through the protocol's indices. This section shows that deterministic carry is a per-leg drift shift, that it is small in barrier terms for cross-asset pairs, and that the one non-level channel — the kink in the rate curve — is also small at a monthly horizon.

8.1. *Carry as an effective drift*

Let r_s and r_b be the supply rate on S_1 and the borrow rate on S_2, constant over $[0,T]$. Then $V_{1,t} = n_1S_{1,0}e^{X_{1,t}+r_st}$, $V_{2,t} = n_2S_{2,0}e^{X_{2,t}+r_bt}$, and $h_t = h_0 + Z_t + ct$ with net carry $c := r_s - r_b$. The carry-adjusted log-prices $\widetilde{X}_{i,t} := X_{i,t} + r_it$ ($r_1 := r_s$, $r_2 := r_b$) are Kou processes with the same diffusion and jump law; under the price-growth convention the shift is exactly $\mu_i \mapsto \mu_i + r_i$, leaving χ_i and $k\eta_{2,+} < 1$ untouched.

Proposition 8.1 (Carry is an effective drift)**.** *The carry-adjusted book coincides with the book of Sections 2–5 under $(\mu_1,\mu_2) \mapsto (\mu_1 + r_s, \mu_2 + r_b)$. Survival and every killed moment $\mathbb{E}[\widetilde{\Pi}_T^k]$, $k \in \mathscr{K}$, are given by the formulas of Sections 4–5 at the shifted drifts.*

The first-passage problem is that of $\widetilde{Z} = Z + ct$ below $-h_0$, and $\widetilde{\Pi}_T$ is (2) with X_i replaced by $\widetilde{X}_i$. Positive c raises survival at fixed h_0; whether by an amount that matters is a quantitative question.

8.2. *The cross-section: when deterministic carry matters*

We screen the Aave v3 Ethereum universe by a rule fixed in advance — blue-chip majors and canonical wrappers, not frozen or paused, listed for the full window, ranked by USD TVL *at the window start*, collateral legs with positive origination LTV, debt legs with borrowing enabled — and report *every* admissible ordered pair: eight reserves, 37 pairs (Appendix B). The diagnostic is the number of barrier standard deviations that carry is worth over the horizon,

$$\mathrm{CB} := \frac{c\,T}{\bar{\sigma}_Z\sqrt{T}}, \tag{35}$$

where $\bar{\sigma}_Z$ is the *realised* total (diffusion plus jump) volatility of Z, measured on daily returns; CB is independent of h_0. We fix $T = 1/12$, long enough for carry to accrue yet short enough for the static assumption of Remark 2. The sampling interval matters. Oracle-implied prices carry microstructure noise at hourly frequency (Appendix A.3), and for near-pegged pairs that noise is most of the hourly variance: hourly sampling puts $\bar{\sigma}_Z$ for wstETH/WETH at 0.39, daily at 0.08.

Collateral	Debt	$\bar{\sigma}_Z$	$r_s - r_b$	CB	debt $U > U_{\mathrm{opt}}$
Stablecoin pairs					
USDT	USDC	0.0039	−1.42%	−1.043	10.4%
USDC	USDT	0.0039	−0.43%	−0.318	8.5%
USDT	USDe	$<10^{-4}$	−0.51%	—	5.2%
Same-underlying pairs					
wstETH	WETH	0.078	−2.51%	−0.093	13.6%
weETH	WETH	0.176	−2.51%	−0.041	13.6%
WBTC	cbBTC	0.073	−0.30%	−0.012	0.0%
Cross-asset pairs					
WETH	USDC	0.514	−2.67%	−0.015	10.4%
WETH	WBTC	0.238	+1.52%	+0.018	0.0%
WBTC	WETH	0.238	−2.51%	−0.030	13.6%
cbBTC	USDC	0.375	−4.54%	−0.035	10.4%

Table 6: Representative rows of the 37-pair cross-section, 2026-03-17 to 2026-08-04, $T = 1/12$. $\bar{\sigma}_Z$: annualised realised volatility of the daily log price ratio; CB: carry-to-barrier ratio (35); last column: share of hours the debt reserve's curve-implied utilization exceeds its kink (Section 8.3). The USDT/USDe oracle ratio does not move at daily resolution, so CB is undefined.

Three groups emerge (Table 6). *For every cross-asset pair, deterministic carry is negligible*: $|\mathrm{CB}| \leq 0.035$, a few percent of one barrier standard deviation over a month, so it cannot change a sizing decision. *Same-underlying pairs* (a liquid-staking token against WETH, or two BTC wrappers) are near-pegged, with $\bar{\sigma}_Z$ between 0.07 and 0.18. There carry is two to three times more material, up to $\mathrm{CB} = -0.09$ for wstETH/WETH, though still under a tenth of a barrier standard deviation; their liquidation risk lies mainly in depegs (Section 8.4). *Stablecoin pairs* collapse $\bar{\sigma}_Z$ to below 0.004, and there carry reaches a full barrier standard deviation ($\mathrm{CB} = -1.04$ for USDT/USDC). The barrier then sits tens of standard deviations away, so liquidation risk is carried by jumps and depegs; for USDT/USDe the oracle ratio does not move at all at daily resolution, and the barrier cannot be reached by price.

A second regularity concerns sign: *a leveraged long-crypto book generally pays carry*. Every pair with crypto collateral and stablecoin or ETH debt has net carry between −1.8% and −4.5%. Wrapped-BTC debt is the exception: those reserves sit at utilization 0.03–0.05 and charge a near-floor rate, so the carry is near zero for staking-token collateral and positive for WETH collateral (WETH/WBTC +1.52%, WETH/cbBTC +1.56%); even there $\mathrm{CB} \approx +0.02$.

8.3. *Regime-switching carry and the* 6×6 *resolvent*

With the rate *level* ruled out, what remains is its *shape*. Aave's variable borrow rate is a deterministic, piecewise-linear function of reserve utilization U,

$$r_b(U) = \begin{cases} r_0 + s_1\, U/U_{\text{opt}}, & U \le U_{\text{opt}}, \\ r_0 + s_1 + s_2\,(U - U_{\text{opt}})/(1 - U_{\text{opt}}), & U > U_{\text{opt}}, \end{cases} \tag{36}$$

with parameters published on-chain. Two consequences follow. First, the curve is invertible, so the observed rate is a sufficient statistic for utilization. This matters because the naive borrowed/supplied ratio is badly biased: for USDC it reports mean utilization 0.881 and a maximum above one, against curve-implied 0.677 and 0.931, and so misclassifies 25.4% of hours as stressed against a true 10.4%. Second, the slope jumps at U_{opt} (published s_2 of 0.60–0.80 for WETH and the major stablecoins and 3.00 for wrapped BTC, against s_1 of 0.027–0.055), so the borrower is short a call on utilization. Fitting a smooth affine diffusion to this kinked transform of a bounded state manufactures spurious jumps and mean reversion.

We therefore let net carry switch between a normal and a stressed state following a two-state Markov chain, with rates (q_1, q_2) and state carries (c_1, c_2) calibrated by thresholding utilization at U_{opt}. Only the drift switches, so this is Proposition 8.1 applied state by state. The barrier system grows from 3×3 to 6×6 and reduces to two decoupled single-regime copies at zero switching (Appendix C, Proposition C.1).

Thresholding the WETH debt leg's hourly utilization at $U_{\text{opt}} = 0.90$ gives a stressed fraction of 19.9% and mean stressed spells of 134 hours ($q_1 = 16.3$, $q_2 = 65.5$ per year), with mean borrow rates of 2.1% (normal) and 4.0% (stressed), so $c_1 = -2.1\%$ and $c_2 = -4.0\%$. This uses the raw series; the curve-implied measure gives 13.6% stressed and would shrink the effect further. Starting from price-only drifts and adding carry back (Table 7), the deterministic mean carry $c = -2.5\%$ lowers survival by 0.8 percentage points near the barrier. The opening regime adds at most 0.1 point, and averaging the two opening states with their stationary weights reproduces the deterministic mean carry to within 0.001 point. The regime channel is *inert at a monthly horizon*: stressed spells of a few days wash out over a month. The 0.8-point cost of mean carry, which CB ≈ -0.03 does not reveal, also shows that CB understates carry's influence near the barrier, where survival is most sensitive to drift. The mechanism is not wrong, only small here: WETH and the stablecoins spend 5–14% of hours above their kinks and would show more at longer horizons or with steeper stressed rates, while wrapped-BTC debt (utilization 0.03–0.05 against $U_{\text{opt}} = 0.45$) has no rate channel at all.

h_0	price-only	det. mean carry	open normal	open stressed
0.10	0.4957	0.4879	0.4882	0.4870
0.25	0.9524	0.9505	0.9505	0.9503
0.50	1.0000	1.0000	1.0000	1.0000

Table 7: Survival at calibrated regime parameters ($q_1 = 16.3$, $q_2 = 65.5$, $c_1 = -2.1\%$, $c_2 = -4.0\%$), benchmark WBTC/WETH dynamics, $T = 1/12$. All runs start from price-only drifts (APRs of Table 2 removed). "Det. mean carry" applies the stationary $c = -2.5\%$ (Proposition 8.1); "open normal/stressed" start the chain in each state. Script `regime_calibration.py`.

Remark 6 (The kink as a convexity). With $r_b(U)$ convex and U random, Jensen's inequality gives

$$\mathscr{J} \;=\; \mathbb{E}\big[r_b(U)\big] - r_b(\mathbb{E}[U]) \;>\; 0 \tag{37}$$

whenever U has mass on both sides of U_{opt}. A deterministic carry evaluated at the mean utilization would set $\mathscr{J} = 0$. The carry used here is the realised average rate, which averages r_b over the observed

utilization path and therefore already contains $\mathscr{J}$. What no deterministic carry can represent is the dependence on the state in which the position opens and on when stress arrives; that is what the regime model adds, and Table 7 shows it is small at a monthly horizon.

8.4. Collateral-depeg sensitivity

A depeg that haircuts collateral by a fraction d scales H_0 by $(1-d)$, so it is the same book at cushion $h_0+\log(1-d)$, re-evaluated on the existing map (7). It matters most for the stablecoin and same-underlying pairs the cross-section flags. Their low $\bar{\sigma}_Z$ makes a thin buffer look safe, and that apparent safety is what permits high leverage. A haircut of a few percent is many times the spread's one-month standard deviation for stablecoin pairs ($\bar{\sigma}_Z\sqrt{T}\approx 0.1\%$) and comparable to it for same-underlying pairs (2–5%), so the leverage that low $\bar{\sigma}_Z$ makes affordable is exactly what makes the book depeg-fragile. We treat the depeg as a scenario overlay so the base dynamics stay pure price. A full treatment would let it co-move with the regime channel, since the deleveraging that breaks a peg also drives utilization through the kink.

9. Sizing on the Map: Illustration, Realised Evidence and an Extension

This section applies the evaluation map to the sizing problem of Section 2.2. Its purpose is to show what the map supports, not to recommend a buffer. The answer turns on a relative-drift input that the data estimate poorly. A full treatment of that problem belongs to a separate exercise, which Section 9.4 outlines.

9.1. Break-even drift views

Write $x := n_2 S_{2,0} = b/(e^{h_0}-b)$ for the debt per unit of equity, so that $x=0$ is the unlevered position and $L_0 = 1+x$. Leverage improves on the unlevered position exactly when the first unit of debt does, and that is a closed-form question. As $x\to 0$ the buffer becomes infinite and the liquidation probability vanishes, so $W_T = (1+x)e^{X_{1,T}} - x e^{X_{2,T}}$ to first order, and

$$\left.\frac{\mathrm{d}\Phi}{\mathrm{d}x}\right|_{x=0} = \left(\mathbb{E}e^{X_{1,T}} - \mathbb{E}e^{X_{2,T}}\right) - \alpha\,\mathrm{Cov}\left(e^{X_{1,T}},\, e^{X_{1,T}} - e^{X_{2,T}}\right), \qquad \left.\frac{\mathrm{d}\,\mathbb{E}[\log W_T]}{\mathrm{d}x}\right|_{x=0} = 1 - \mathbb{E}\,e^{X_{2,T}-X_{1,T}}, \tag{38}$$

for mean–variance (6) and for Kelly. All terms are joint exponential moments $e^{T\Psi_{\mathrm{M}}(a_1,a_2)}$. The *break-even drift view* $\nu^\star$ is the spread drift $\nu = \mu_1 - \mu_2$ (price growth plus net carry) at which the derivative vanishes, holding the level μ_2 at its sample value. Without liquidation both objectives are concave in x (mean–variance is a concave quadratic, log a concave function of an affine one), so the sign of the derivative at $x=0$ decides between the unlevered position and some leverage. Liquidation breaks this argument, so we check it numerically: a search over 30 debt levels spanning the full buffer range, from $x=10^{-3}$ ($H_0\approx 781$) to the origination maximum, with exact recovery and the floored Kelly objective of Section 7.5, confirms the switch at $\nu^\star$ for every objective and calibration.

For Kelly the condition $\mathbb{E}e^{X_{2,T}-X_{1,T}} = 1$ is $\Psi_{\mathrm{M}}(-1,1)=0$, which does not depend on T and, with independent jump processes, solves in closed form:

$$\nu^\star_{\mathrm{Kelly}} = \underbrace{\sigma_1^2 - \rho\sigma_1\sigma_2}_{\text{diffusion}} + \underbrace{\lambda_1\left[M_1(1)+M_1(-1)-2\right]}_{\text{WBTC jumps}}, \qquad M_1(s) = \frac{p_1}{1-s\eta_{1,+}} + \frac{1-p_1}{1+s\eta_{1,-}}. \tag{39}$$

The break-even views are small (Table 8): about 0.4–1.6% per year under the headline calibration and 2.4–11% under the full-history ones, close to linear in α, with Kelly near $\alpha = 1$. Equation (39) says why. The diffusion part is negative, because WBTC is the less volatile leg ($\sigma_1 < \rho\sigma_2$): at the margin, shorting WETH against WBTC reduces diffusive risk. The break-even is positive only through WBTC's own jumps, which the short leg does not hedge: $-0.004 + 0.012$ under the headline calibration, $-0.017 + 0.065$ under the full-history daily one and $-0.009 + 0.054$ under the 2024 one. The second term rests on the jump parameters, which the data identify weakly (Appendix A), and a common jump component, which the model excludes, would reduce it by the covariance of the two legs' jumps. Section 9.3 shows that this is the case in the data: the realised break-even of WBTC/WETH is negative. With no relative-drift view at all, ν equals the net carry ($\approx -2.5\%$), below every break-even, so the unlevered position is optimal for every objective. Above $\nu^\star$ the optimal leverage rises quickly with the view (Figure 2).

Calibration	model $\bar{\sigma}_Z$	$\alpha = 0.5$	$\alpha = 1$	Kelly	$\alpha = 2$
Headline, 4h, 2026-03 to 08 (0.38 y)	0.335	+0.004	+0.008	+0.008	+0.016
Full history, daily (3.7 y)	0.478	+0.027	+0.054	+0.048	+0.110
2024 onward, 4h (2.8 y)	0.485	+0.024	+0.049	+0.045	+0.099

Table 8: Break-even spread-drift views $\nu^\star$ (per year, price growth plus net carry): the unlevered position is optimal below $\nu^\star$ and some leverage above it, for mean–variance on terminal wealth with exact recovery and for Kelly, from (38), with μ_2 at the sample level of the matching window (the full-history one for the 2024 calibration). "Model $\bar{\sigma}_Z$" is the calibrated total (diffusion plus jump) spread volatility. Script `breakeven_drift.py`.

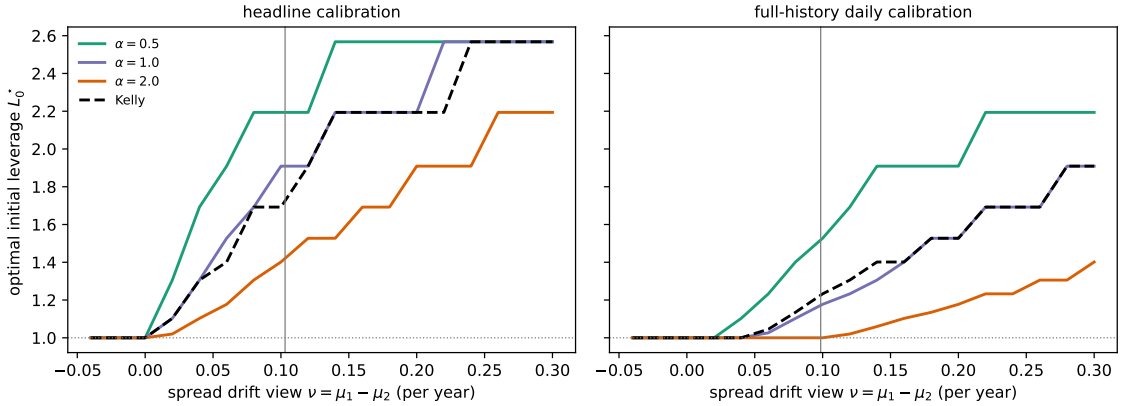


Figure 2: Optimal initial leverage $L_0^\star$ against a drift view $\nu = \mu_1 - \mu_2$, mean–variance on terminal wealth ($\alpha = 0.5, 1, 2$) and Kelly, exact recovery, over the full buffer range. Left: headline calibration; right: full-history daily calibration. The optimum is unlevered ($L_0^\star = 1$) below the break-even view and rises quickly above it. The vertical line marks the sample drift of each window. The optimum is located on 30 debt levels plus the unlevered position, hence the steps. Script `breakeven_drift.py`.

9.2. What the data say about the drift

The spread drift has two parts, $\nu = \nu_{\text{price}} + c$. The net carry $c = r_s - r_b$ is observed on-chain, about -2.5% per year for WBTC/WETH. The relative price growth ν_{price} is the expected outperformance of WBTC over WETH, and it is estimated poorly at any sampling frequency, because the standard error of

a drift depends only on calendar length (Merton, 1980). The ECF drift normalizers of Table 2 are fitted to de-meaned returns and are not estimates of it, so we use sample means. The headline window gives $\nu = +0.10$ (price $+0.13$, carry -0.025) with a Newey–West standard error of 0.34; the full 3.7-year daily history gives $\nu = +0.10$ (price $+0.13$, carry -0.03) with a Newey–West and block-bootstrap standard error of 0.19; and the yearly estimates of ν_{price} range from -0.16 to $+0.34$ with standard errors of 0.3–0.5. Both point estimates lie above every headline break-even and above most full-history ones, and at them the optimal leverage is moderate (1.4–2.2 under the headline calibration, 1.0–1.5 under the full-history ones). But the standard errors are several times the gap between estimate and break-even: telling $+0.10$ from the full-history Kelly break-even of $+0.05$ at two standard errors would need a standard error of 0.025, about two centuries of data. Both reserves accept collateral and borrowing, so the book can also be held the other way round. Reversing the legs flips the sign of ν_{price} and changes the carry: over the full history, WETH collateral against WBTC debt has $\nu_{\text{price}} = -0.13$ and $c = +1.1\%$, so $\nu = -0.11$. Its Kelly break-even, the mirror image of (39) with the legs exchanged, is $+0.10$ to $+0.19$ across the calibrations and has a positive diffusion part, because WETH is the more volatile leg; the reversed book is thus far from paying, even though WETH's higher liquidation threshold (0.83 against 0.78) would allow more leverage. A relative-drift view therefore chooses the orientation as well as the buffer.

A carry-anchored Bayesian layer turns this into a probability. Take $\nu \sim \mathcal{N}(c, 0.20^2)$, the view that relative prices have no predictable drift, and update it with the sample estimate. The posterior probability that leverage pays, $\mathbb{P}(\nu > \nu^\star)$, is 48–51% for every objective under the headline calibration and 30–54% under the full-history ones, treating $\nu^\star$ as known. Under a flat prior, which takes the sample estimate at face value, it is 60–62% and 48–65%. Either way the data leave the question close to open. Maximising the posterior expectation of $\mathbb{E}[\log W_T]$ (Bayes–Kelly) has an exact rule at the margin: with posterior $\mathcal{N}(m, s^2)$, lever if and only if $m > \nu^\star_{\text{Kelly}} + \frac{1}{2} T s^2$. The variance term is small (0.001 at the headline posterior), so the rule is decided by the posterior mean. The carry-anchored prior pulls it to $+0.008$ (headline) and $+0.039$ (full history), just below the thresholds $+0.009$ and $+0.046$–0.049, and Bayes–Kelly selects the unlevered position under all three calibrations. Under the flat prior $m = \hat{\nu} \approx 0.10$ and it levers. The unlevered choice therefore reflects the prior's shrinkage toward zero relative drift, not estimation risk as such. Out of sample, the fitted ECF drift also worsens the map's liquidation forecasts relative to setting $\nu_{\text{price}} = 0$ (Appendix A.3).

9.3. Did leverage pay? A realised backtest

The break-even rule says when leverage pays; whether it did is a question for realised data, and one that needs no model. We replay every pair of the cross-section (Section 8.2; 37 pairs) hour by hour on the Aave v3 Ethereum history: oracle-implied prices, the collateral's supply rate and the debt's borrow rate. Windows run from the first hour at which both reserves are live (27 January 2023 for the original reserves, up to 23 September 2024 for cbBTC) to 13 October 2026. Equity is in dollars. At each roll date the book is reset to initial leverage L_0; between rolls interest accrues on both legs, and whenever the health factor falls below one at an hourly mark a keeper repays half the debt (all of it below 0.95) and seizes collateral worth the repayment plus the reserve's bonus, as on Aave v3. The benchmark holds the collateral unlevered. Rolled books are averaged over four roll phases and buy-and-hold books over four start dates 120 days apart. Standard errors are Newey–West on daily excess log returns (60-day lag).

Leverage rarely paid (Table 9, Figure 3). At $L_0 = 1.25$ it beat the unlevered book in 4 of 37 pairs — WBTC against USDC, USDT and WETH, and cbBTC against WETH, all books long BTC in a window in which BTC rose 3.6-fold and doubled against ETH — and in none significantly. Of 1,332 pair, leverage and horizon combinations, one has $t > 2$ and 338 have $t < -2$; the four stablecoin pairs lose significantly at every leverage up to $3\times$ and every horizon, because their net carry over the window is negative and nearly deterministic. Losses grow quickly with leverage, through volatility drag at every

Roll horizon	$L_0 = 1.25$	$L_0 = 1.5$	$L_0 = 2$	$L_0 = 3$
1 day	−0.038 (4)	−0.102 (4)	−0.287 (3)	−0.768 (0)
30 days	−0.042 (4)	−0.113 (4)	−0.331 (4)	−0.994 (1)
365 days	−0.008 (5)	−0.112 (5)	−0.559 (3)	−0.889 (2)
Buy and hold	−0.017 (12)	−0.036 (12)	−0.303 (10)	−0.451 (7)

Table 9: Realised excess log growth per year of the levered over the unlevered book: median over the 37 pairs, with the number of pairs in which leverage paid in parentheses. Aave v3 liquidation rules, current reserve parameters, no trading costs. Script `realised_backtest.py`.

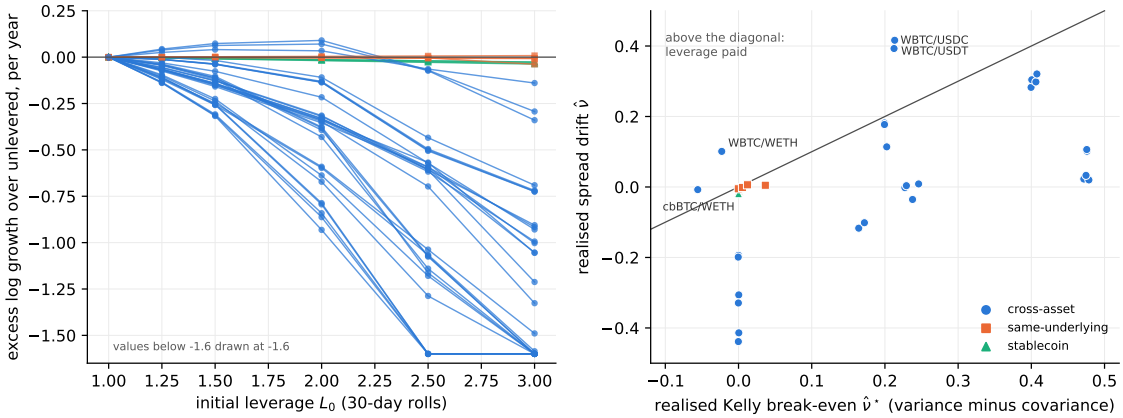


Figure 3: Left: realised excess log growth against initial leverage, 30-day rolls, one line per pair. Right: realised spread drift $\hat{\nu}$ against the realised Kelly break-even $\hat{\nu}^\star$, both from daily gross returns including interest; above the diagonal a little leverage paid. Script `fig_realised_backtest.py`.

horizon and through liquidations once the book is left for a week or more: from $L_0 = 2.5$ most pairs are liquidated at least once at those horizons, and a book rolled at the maximum LTV loses in every pair. A 10 bp trading cost per reset leaves these numbers almost unchanged, and so does the paper's full-liquidation convention for rolled books up to $L_0 = 2$; for buy-and-hold at high leverage it loses less than Aave's partial liquidations, because after one liquidation the residual is held unlevered. Rolling more often does not help: at low leverage the excess is about the same from one day to a year. Buy and hold does better, because a static book's leverage falls as it gains and it avoids the volatility drag of re-levering, but it still loses in most pairs. Nor is short-term profitability a matter of calendar time: at $L_0 = 2$ and 30-day rolls, 17% of cross-asset pair-years were positive, almost all of them BTC-long books in 2023–24 or ETH-long books against dollars in 2023, and none of the 29 in 2025. The window is one cycle — a bull market to the 2025 peaks, then a decline of 53% for BTC and 68% for ETH to mid-2026 — with one dominant cross-asset trend, so these numbers describe this history, not a law. A simple ex-ante rule that levers to 2× for a month only after a positive trailing-year spread return lost less than always levering in 29 pairs but beat the unlevered book in only 5.

The realised break-even confirms the rule and exposes the model. For a daily-rolled book the marginal log growth of the first unit of debt is $\mathbb{E}[1 - R_2/R_1]$ in daily gross returns, so leverage pays at the margin if and only if $\hat{\nu} > \hat{\nu}^\star$, the realised counterpart of (39); this is a first-order identity, and it sorts all 37 pairs correctly. Its value lies in the comparison with the model. For WBTC/WETH, $\hat{\nu}^\star = -0.023$ (daily; -0.026 four-hourly), against the full-history model's $+0.048$. The legs' realised correlation is 0.81, and on 85% of the days with a three-standard-deviation move in either leg both legs move by more than one

and a half standard deviations in the same direction; the calibrated model, whose jumps are independent, has correlation 0.50. With only Brownian correlation to carry co-movement, the fit cannot match the common jumps: it understates both legs' variances (WBTC 0.15 against 0.21 per year) and, by much more, their covariance (0.11 against 0.24), so the spread variance is overstated (0.23 against 0.14) and the break-even, collateral variance minus covariance, turns positive. The same overstatement inflates the model's liquidation probabilities and is consistent with their excess over realised frequencies out of sample (Appendix A.3). A common jump component — a third Poisson process moving both legs, with double-exponential marks so that the spread's jumps stay in the mixed-exponential class of Section 5, at the cost of more phases and a larger barrier system — is therefore the most consequential extension of the model.

9.4. A decision-theoretic extension

These results suggest a separate treatment of the sizing problem, which we leave to future work. Its question is whether a leveraged position of this kind can be sized from data at all. Four pieces would be needed. *(i)* A preference-free break-even: the killed law of Z_T and the overshoot law of Proposition 7.1 determine the full distribution of W_T, so one can find the drift views below which the unlevered position second-order stochastically dominates every levered one. Below that view every risk-averse investor agrees, not only the objectives above. *(ii)* The CRRA family, which extends Section 7.5 by the binomial series in place of the logarithmic one, with the same tilted moments. *(iii)* An explicit account of estimation risk in the tradition of Barberis (2000), Kan and Zhou (2007) and DeMiguel et al. (2007), and of fractional Kelly (MacLean et al., 2011), with the break-even view $\nu^\star$ as the decision rule: lever only if one's relative-drift view, including carry, exceeds it, and size the position by the posterior rather than the point estimate. *(iv)* A repeated, monthly-rolled version in which liquidation is partial ruin and long-run growth is the natural criterion.

10. Conclusion

The paper develops a semi-analytic evaluation map for leveraged DeFi long–short positions under liquidation and jump risk. A change of numeraire reduces the two-asset barrier problem to one dimension, and the Kou structure keeps every admissible moment, the survival probability and the exact liquidation recovery within one 3×3 barrier system. Bivariate tilts of that system give the overshoot law and the exact probability of bad debt, and with them the expected log of terminal wealth. Protocol interest enters as a drift shift, and its rate-curve convexity through a 6×6 resolvent. The map is noise-free, fast, and validated against an accuracy-matched simulator. Empirically, its one-month liquidation probabilities lie above realised frequencies out of sample, systematically so for some predictors, and two of its inputs depend materially on how the data are sampled: realised spread volatility at hourly frequency is inflated by oracle noise, and the size of the jump-overshoot correction depends on the calibration window.

Applied to sizing, the map gives a closed-form break-even drift above which leverage pays. Under the calibrated model it is about 0.4–11% per year depending on risk attitude and calibration. The sample drift lies near or above it, but with standard errors several times the gap, so the data cannot tell an investor whether to lever. Ex post, leverage rarely paid: replayed under Aave's liquidation rules over 2023–2026, it beat the unlevered position in 4 of 37 pairs at modest leverage and in none significantly, and higher leverage lost at every holding horizon, through volatility drag and, for books left a week or more, liquidations. The realised break-evens sort the pairs correctly but disagree with the model's for WBTC/WETH, because the model's independent jumps cannot reproduce the legs' common jumps. The map is therefore best used to evaluate a buffer under a stated view or stress scenario, rather than

to calibrate one, and its most consequential extension is a common jump component. The model also abstracts from oracle delays, liquidation execution and dynamic collateral management. Beyond the decision-theoretic programme of Section 9.4, further extensions are an endogenous utilization process in place of the exogenous regime chain, the stochastic control problem of Remark 2, and a market-impact channel that makes the liquidation bonus ℓ endogenous in the overshoot regime.

Appendix A: Empirical Calibration

A.1. Estimator

The model characteristic function over an observation interval Δ is available in closed form,

$$\phi_\theta(u,v;\Delta) = \exp\big(\Delta\Psi_\theta(u,v)\big). \tag{40}$$

Given N bivariate log-returns $(r_{1,n}, r_{2,n})$, with empirical characteristic function

$$\widehat{\phi}_N(u,v) = \frac{1}{N}\sum_{n=1}^{N} \exp\big(i(u\,r_{1,n} + v\,r_{2,n})\big), \tag{41}$$

we estimate θ by penalised ECF matching,

$$\begin{aligned}\widehat{\theta} = \arg\min_{\theta\in\Theta} &\sum_{(u,v)\in\mathscr{G}} w(u,v)\,\big|\widehat{\phi}_N(u,v) - \phi_\theta(u,v;\Delta)\big|^2 \\ &+ \kappa_\lambda \sum_{i=1}^{2}\big(\log\lambda_i - \log\bar{\lambda}_i\big)^2 + \kappa_\eta \sum_{i=1}^{2}\sum_{\pm}\big(\log\eta_{i,\pm} - \log\bar{\eta}_{i,\pm}\big)^2.\end{aligned} \tag{42}$$

The first term is the ECF distance of Singleton (2001), which can attain ML efficiency over a rich grid (Feuerverger and McDunnough, 1981) and avoids a two-dimensional density inversion inside every optimizer call (for univariate DEJD likelihood estimation see Ramezani and Zeng 2007). The two soft penalties, with $\kappa_\lambda = 0.05$ and $\kappa_\eta = 0.02$, anchor the intensities to a two-moment (variance and fourth-cumulant) estimate $\bar{\lambda}_i$ and the jump sizes to peak-over-threshold mean excesses $\bar{\eta}_{i,\pm}$. They exist to stop the optimiser sliding along the λ–η ridge discussed in Appendix A.2 toward very many, very small jumps. The fitted jump parameters are therefore partly anchored rather than purely ECF-determined, which should be kept in mind when reading them.

Grid. Frequencies are standardised: a dimensionless base value a maps to the raw frequency a/s, where s is the robust (IQR) scale of the relevant return series, so the same grid serves any sampling interval. Six groups are used: marginal directions for each asset (eleven base values from 0.1 to 10), same-sign and opposite-sign joint directions on a 6×6 grid ($a,b \in \{0.25, 0.5, 1, 2, 3, 5\}$) that identify ρ, the spread direction $(s,-s)$ scaled by the robust scale of Z ($a \in \{0.1,\ldots,5\}$), which matches the characteristic function of the liquidation-driving process directly, and a jump-tail group at high base frequencies ($a \in \{8, 12, 16, 20, 25\}$) on the marginal and spread directions. Each point enters with its conjugate, giving 234 points for 13 parameters. Weights are $w = \exp(-0.02(a^2+b^2))$ in the base frequencies, doubled on the spread group.

Optimisation. Constraints are enforced by reparameterisation ($\sigma_i, \lambda_i, \eta_{i,-}$ via exp; $p_i, \eta_{1,+}$ via sigmoid; ρ via tanh), with $\eta_{2,+} = \eta_{2,+}^{\max}\,\mathrm{sigmoid}(d_2)$, $\eta_{2,+}^{\max} = (1-\varepsilon)/K$, so that $K\eta_{2,+} < 1$ holds by construction; the headline bounds are $\lambda_i \le 5000$, $p_i \in [0.001, 0.999]$ and $\eta \ge 10^{-5}$. The non-convex objective is

warm-started by a threshold decomposition in the spirit of Mancini (2009): returns far from the median in robust-scale units are flagged as jumps, σ_i and ρ come from the remaining observations, and μ_i from the sample drift with a damped mean-jump correction. L-BFGS-B is then run from a multi-start cloud (30 starts for the headline fit): the warm start, a set of ridge-traversing starts that scale λ by c and η by $1/\sqrt{c}$, and randomly perturbed scenario starts. The lowest objective is kept. The implementation is the `optimal_long_short.calibration` package; `jobs/calibrate_kou.py` demonstrates parameter recovery on synthetic data.

A.2. Identifiability and model comparison

Jump-diffusions have approximate ridges along directions preserving total jump variance $\lambda_i(p_i\eta_{i,+}^2 + (1-p_i)\eta_{i,-}^2)$ (Ramezani and Zeng, 2007). A circular block bootstrap (block length 12, $B = 200$, each resample recalibrated with the headline bounds and penalties from 8 starts; Table 10) shows the consequences. Diffusion volatilities and correlation are sharp (relative SEs of about 6% for σ_i and 2% for ρ). The drifts and the jump parameters are not: $\hat{\mu}_1 = 0.30 \pm 0.72$, $\hat{\mu}_2 = 1.16 \pm 1.17$, and their difference -0.86 has SE 1.06. These are the ECF drift normalizers, not sample drifts (Section 6), so their standard errors measure how loosely the ECF pins the drift parameters rather than the precision of a drift estimate. The intensities carry relative SEs near 30%, even with the soft anchors of (42) holding them toward the moment estimates, and the jump-size intervals span a factor of five to six for $\eta_{i,+}$ and about fifteen for $\eta_{i,-}$. The block length of two days is short relative to the mean reversion visible at a week in this window, so these standard errors may understate the uncertainty. No resample reaches the $\lambda \leq 5000$ bound. The data pin down σ_Z, not the drift or the jump decomposition.

Parameter	point	bootstrap SE	95% CI
σ_1	0.264	0.016	[0.228, 0.295]
σ_2	0.328	0.019	[0.296, 0.370]
ρ	0.847	0.020	[0.800, 0.878]
μ_1	0.302	0.725	[−0.61, 1.75]
μ_2	1.161	1.172	[−0.91, 3.58]
$\mu_1 - \mu_2$	−0.858	1.058	[−2.65, 1.36]
λ_1	358	106	[219, 628]
λ_2	451	134	[253, 770]
p_1	0.770	0.210	[0.24, 1.00]
$\eta_{1,+}$	0.0041	0.0036	[0.0015, 0.0085]
$\eta_{2,+}$	0.0113	0.0041	[0.0031, 0.0191]

Table 10: Block-bootstrap standard errors ($B = 200$), each resample recalibrated with the headline bounds ($\lambda \leq 5000$) and anchoring penalties. The two drift estimates are correlated (+0.46). Script `jobs/bootstrap_se.py`.

In the headline window the fitted jumps are tail-fattening, not rare and large. Fitting GBM, Merton (Gaussian jumps) and Kou to the ECF of the spread Z on a common grid, Kou has the smallest in-sample objective (1.7×10^{-6} against 1.3×10^{-5} and 3.4×10^{-4}), is preferred under a BIC-style penalty, and matches the empirical 1%/99% one-step quantiles (−1.59%/+1.43% against −1.52%/+1.42%). But it reproduces the excess kurtosis of about 3 with *many small* jumps — about 2,700 per year of mean size 0.2–0.4% in this univariate fit, and 350–450 per year of 0.4–1.1% in the bivariate headline fit — not with rare large moves; flash crashes enter only through the scaled experiments of Section 7.3 and the full-history calibration below. Out of sample (fit on the first 60%, score on the last 40%), Kou's edge over Merton all but vanishes (1.13 against 1.14×10^{-3}; GBM 1.60×10^{-3}).

A.3. Full history, sampling frequency and out-of-sample validation

Data. The hourly snapshots extend back to the launch of Aave v3 on Ethereum, giving 3.7 years (2023-01-27 to 2026-10-13). A snapshot exists only in hours with market activity, so the thin early WBTC market has gaps: 68.5% of 2023 hours carry a real snapshot, against 96% in 2024 and essentially all hours since. Forward-filled prices would create spurious zero returns and catch-up jumps, so each bar takes the last *real* price within it, and returns that span a bar without a snapshot are dropped (146 of about 8,100 four-hourly returns, none of the daily ones).

Oracle noise and the sampling horizon. Hourly returns of the oracle-implied spread have lag-one autocorrelation of -0.16 to -0.25 depending on the period, the signature of microstructure noise (Aït-Sahalia et al., 2005), which inflates hourly volatility by up to about one half. This is why the cross-section of Section 8.2 uses daily returns, and why the calibrations stop at four hours. Even at four hours some of it may remain: the headline model's total spread volatility (0.335) exceeds the daily realised value over the same window (0.238), so part of the fitted jump variance may be noise or short-horizon mean reversion, and part reflects the independent-jump restriction discussed in Section 9.3. Relative to four-hourly returns, the variance ratios (Lo and MacKinlay, 1988) over the full sample are 0.86 at one day, 0.88 at one week and 0.92 at one month, close to the independent increments the Lévy model assumes. They vary by period, however: 0.33 at one month in 2023, 1.56 in 2025, and 0.54 at one week in the headline window, which is unusually mean-reverting.

Full-history calibrations. Table 11 compares the headline fit with the same pipeline applied to daily bars over the full history and to four-hourly bars from 2024. Both longer samples give a larger total spread volatility. The daily fit trades many small jumps for fewer, larger ones ($\lambda \approx 85$ per year, $\eta_{2,+} \approx 3.2\%$), which is what raises the overshoot correction to -11% and the bad-debt probability to 3×10^{-4} (Section 7.4). The ECF drift normalizer $\mu_1 - \mu_2$ ranges from -0.86 to $+0.43$ across the three; it is fitted to de-meaned returns and is not an estimate of the sample drift, which Section 9 uses instead.

Calibration	σ_Z	model $\bar{\sigma}_Z$	ρ	λ_1	λ_2	$\eta_{2,+}$	ECF $\mu_1 - \mu_2$
Headline, 4h, 0.38 y	0.175	0.335	0.847	358	451	0.011	-0.86
Full history, daily, 3.7 y	0.196	0.478	0.884	90	81	0.032	-0.37
2024 onward, 4h, 2.8 y	0.235	0.485	0.794	397	277	0.015	$+0.43$

Table 11: WBTC/WETH calibrations with the headline pipeline (EWM de-meaning, penalised ECF, APR drift adjustment). σ_Z: diffusive spread volatility; model $\bar{\sigma}_Z$: total including jumps. Script `jobs/horizon_analysis.py`.

Liquidation probabilities, in and out of sample. The headline window was calm for the spread: rolling one-month windows in it crossed -0.07 only 2% of the time, while from 2024 onward they did so 36% of the time (47%, 23% and 12% for $h_0 = 0.05$, 0.10 and 0.15). Over 2024 onward the three calibrations match these frequencies to within about a factor of two, in either direction. The decisive test is out of sample (Table 12). For each of 33 months from 2024-01, the model is calibrated only on data before the month, predicts the probability that the spread falls by h_0 within it, and is scored against the realised four-hourly path. The model monitors the barrier continuously and the data only every four hours, which by itself makes the model's probabilities higher than the observed frequencies; part of the overprediction below is this monitoring effect. Three findings follow. The map's probabilities are on average *higher* than the realised frequency, by a factor of about 1.3–1.6 (two to three at $h_0 = 0.15$).

Treating the 33 months as independent, the over-prediction is 1.0 to 2.8 standard errors depending on predictor and buffer. A moving-block bootstrap over months, which respects their serial dependence, gives 95% intervals for the bias that exclude zero for the four-hourly carry-only predictor at every buffer, but not for the daily fitted-drift one. The bias is therefore systematic for some predictors and not established for others. The four-hourly monitoring pushes in this direction, and so does the mean reversion of 2024 (one-month variance ratio 0.49); 2025 trended instead (1.56). A third cause is structural: with independent jumps, the calibrated model understates the legs' covariance and so overstates the spread variance, by about 60% over the full history (Section 9.3). Daily calibration does better than four-hourly. And no predictor beats the constant base rate in Brier score, so beyond its upward bias the map has no month-to-month skill: a driftless GBM with trailing volatility does at least as well as Kou, and the fitted drift worsens the Brier score of every Kou forecast.

	mean one-month crossing probability			Brier score
Predictor	$h_0 = 0.05$	$h_0 = 0.10$	$h_0 = 0.15$	$h_0 = 0.10$
Realised frequency	0.455	0.273	0.091	0.198 (base rate)
Kou, daily, 12 months, fitted drift	0.595	0.357	0.203	0.267
Kou, daily, 12 months, carry-only drift	0.620	0.375	0.207	0.219
Kou, 4h, 6 months, fitted drift	0.585	0.386	0.258	0.323
Kou, 4h, 6 months, carry-only drift	0.678	0.436	0.258	0.226
GBM, trailing 30-day volatility	0.627	0.354	0.184	0.213

Table 12: Out-of-sample backtest of one-month liquidation probabilities, WBTC/WETH, 33 months 2024-01 to 2026-09. Each month's predictor uses only earlier data, with the headline pipeline but 8 optimiser starts instead of 30; the model horizon is $1/12$ year and the realised window 30 days. "Carry-only drift" sets the relative price drift to zero and keeps the window's net carry. The base-rate Brier score uses the realised frequency as a constant forecast. Script `jobs/rolling_backtest.py`.

Appendix B: Full cross-section

Table 13 lists all 37 ordered pairs surviving the screen of Section 8.2; columns as in Table 6. For every cross-asset pair $|\mathrm{CB}| \le 0.035$; same-underlying pairs reach 0.09, and among stablecoin pairs USDT/USDC reaches -1.04 while USDC/USDT is -0.32 and USDC/USDe $+0.03$.

Appendix C: Regime-switching carry: the 6×6 resolvent

Let $J_t \in \{1,2\}$ (normal, stressed) be a Markov chain with generator $Q = \left(\begin{smallmatrix} -q_1 & q_1 \\ q_2 & -q_2 \end{smallmatrix}\right)$ and net carry c_i in state i. Only the drift switches, so the exponent of Z in state i is

$$\psi^i(s) = \psi_Z^{(k)}(s) + c_i s, \tag{43}$$

with diffusion, jump law, phase rates and admissibility bounds unchanged. The killed resolvents $\widehat{U}^i$ started in state i satisfy

$$(q + q_1 - \mathscr{L}^1)\widehat{U}^1 - q_1\widehat{U}^2 = g_k, \qquad -q_2\widehat{U}^1 + (q + q_2 - \mathscr{L}^2)\widehat{U}^2 = g_k, \tag{44}$$

and an exponential mode $Bv(\gamma)e^{\gamma x}$ solves the homogeneous system when v spans the kernel of $A(\gamma) = \left(\begin{smallmatrix} q+q_1-\psi^1(\gamma) & -q_1 \\ -q_2 & q+q_2-\psi^2(\gamma) \end{smallmatrix}\right)$, i.e. when $\Delta(\gamma) := \det A(\gamma) = 0$.

Coll.	Debt	$\bar{\sigma}_Z$	carry	CB	str.	Coll.	Debt	$\bar{\sigma}_Z$	carry	CB	str.
USDC	USDT	0.0039	-0.43%	-0.318	8.5%	WETH	cbBTC	0.253	+1.56%	+0.018	0.0%
USDC	USDe	0.0039	+0.05%	+0.034	5.2%	cbBTC	USDC	0.375	-4.54%	-0.035	10.4%
USDC	WBTC	0.370	+3.33%	+0.026	0.0%	cbBTC	USDT	0.374	-4.11%	-0.032	8.5%
USDC	WETH	0.514	+1.17%	+0.007	13.6%	cbBTC	USDe	0.374	-3.63%	-0.028	5.2%
USDC	cbBTC	0.375	+3.37%	+0.026	0.0%	cbBTC	WBTC	0.073	-0.35%	-0.014	0.0%
USDT	USDC	0.0039	-1.42%	-1.043	10.4%	cbBTC	WETH	0.253	-2.51%	-0.029	13.6%
USDT	USDe	$<10^{-4}$	-0.51%	—	5.2%	weETH	USDC	0.524	-4.54%	-0.025	10.4%
USDT	WBTC	0.369	+2.77%	+0.022	0.0%	weETH	USDT	0.523	-4.11%	-0.023	8.5%
USDT	WETH	0.514	+0.61%	+0.003	13.6%	weETH	USDe	0.523	-3.63%	-0.020	5.2%
USDT	cbBTC	0.374	+2.82%	+0.022	0.0%	weETH	WBTC	0.276	-0.36%	-0.004	0.0%
WBTC	USDC	0.370	-4.54%	-0.035	10.4%	weETH	WETH	0.176	-2.51%	-0.041	13.6%
WBTC	USDT	0.369	-4.11%	-0.032	8.5%	weETH	cbBTC	0.274	-0.31%	-0.003	0.0%
WBTC	USDe	0.369	-3.63%	-0.028	5.2%	wstETH	USDC	0.522	-4.54%	-0.025	10.4%
WBTC	WETH	0.238	-2.51%	-0.030	13.6%	wstETH	USDT	0.521	-4.11%	-0.023	8.5%
WBTC	cbBTC	0.073	-0.30%	-0.012	0.0%	wstETH	USDe	0.521	-3.63%	-0.020	5.2%
WETH	USDC	0.514	-2.67%	-0.015	10.4%	wstETH	WBTC	0.246	-0.35%	-0.004	0.0%
WETH	USDT	0.514	-2.24%	-0.013	8.5%	wstETH	WETH	0.078	-2.51%	-0.093	13.6%
WETH	USDe	0.514	-1.76%	-0.010	5.2%	wstETH	cbBTC	0.260	-0.31%	-0.003	0.0%
WETH	WBTC	0.238	+1.52%	+0.018	0.0%						

Table 13: All 37 screened Aave v3 collateral–debt pairs, 2026-03-17 to 2026-08-04, $T = 1/12$. $\bar{\sigma}_Z$ from daily returns of the oracle-implied price ratio; "carry" is $r_s - r_b$; "str." is the share of hours the debt leg's curve-implied utilization exceeds U_{opt}. Source: `results/pair_cross_section_daily.csv`.

Proposition C.1 (Regime-switching killed resolvent)**.** *Both ψ^i share the four poles of $\psi_Z^{(k)}$; writing $\psi^i = N^i/D$ with $\deg D = 4$, $\deg N^i = 6$,*

$$\Delta(s)\,D(s)^2 = \widetilde{N}^1(s)\widetilde{N}^2(s) - q_1 q_2 D(s)^2, \qquad \widetilde{N}^i := (q+q_i)D - N^i,$$

a degree-12 polynomial with, generically, six roots of negative real part. The boundary conditions also number six — absorption and the two downward-phase conditions of Proposition 5.2 in each state (the coupling terms carry no generator and add no zero-extension residual) — so $B \in \mathbb{C}^6$ solves, for $i = 1,2$ and $d = 1,2$,

$$\sum_{m=1}^{6} B_m v^i(\gamma_m) = -\sum_{j=0}^{k} P_j^i e^{-jh_0}, \qquad \sum_{m=1}^{6} B_m v^i(\gamma_m)\frac{r_{d,-}}{r_{d,-}+\gamma_m} = -\sum_{j=0}^{k} P_j^i \frac{r_{d,-}}{r_{d,-}+j} e^{-jh_0},$$

where $A(j)P_j = (c_{k,j}, c_{k,j})^\top$. At $q_1 = q_2 = 0$ this is two decoupled copies of Proposition 5.2.

The kernel vector $v(\gamma)$ should be taken from the adjugate of $A(\gamma)$: in the decoupled limit one adjugate column vanishes while the other selects the state a root belongs to. The non-resonance condition becomes $\Delta(j) \neq 0$.

Lemma C.2 (Generic invertibility)**.** *With the decaying roots labelled analytically in (q_1, q_2) (possible while they remain simple), $\det \mathbf{M}_{\text{bar}}$ is analytic in (q_1, q_2) and at $q_1 = q_2 = 0$ factorises into two single-regime determinants, each non-zero by Lemma 5.3; hence the 6×6 system is invertible off a null set of switching rates.*

We treat invertibility at the calibrated (q_1, q_2) as numerically verified: the condition number rises only from 81 to 844 for real $q \in [0.1, 100]$ (we did not tabulate it on the complex Talbot contour). The solver reproduces the single-regime answer to 10^{-14} when $c_1 = c_2$ with $q_1 \neq q_2$ and to 10^{-15} in the decoupled limit ($k = 0,1,2$), and agrees with a switching Monte Carlo once its discretisation bias is refined away.

Appendix D: Proofs

Proof of Lemma 3.1. Using $\mathbb{E}_{\mathbb{P}}[e^{\alpha X_{1,t}+\beta X_{2,t}}] = e^{t\Psi(-i\alpha,-i\beta)}$,

$$\mathbb{E}_{\mathbb{P}^{(k)}}\left[e^{sZ_t}\right] = e^{-t\Psi(0,-ik)}\,\mathbb{E}_{\mathbb{P}}\left[e^{sX_{1,t}+(k-s)X_{2,t}}\right] = e^{t[\Psi(-is,-i(k-s))-\Psi(0,-ik)]}.$$

The density is exponential-linear in (X_1, X_2), so by the Esscher/Girsanov theorem for Lévy processes Z remains Lévy under $\mathbb{P}^{(k)}$ with exponent $\psi_Z^{(k)}(s) = \Psi(-is,-i(k-s)) - \Psi(0,-ik)$. Substituting (14) with $u^2 = -s^2$, $v^2 = -(k-s)^2$, $uv = -s(k-s)$ and $\varphi_{J_i}(-i\cdot) = M_i(\cdot)$: the drift terms give $(\mu_1^X - \mu_2^X)s$; the diffusion terms give $\frac{1}{2}\sigma_1^2 s^2 + \frac{1}{2}\sigma_2^2(k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) - \frac{1}{2}\sigma_Z^2 k^2 = \frac{1}{2}\sigma_Z^2 s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s$; and the jump terms give $\lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s) - M_2(k))$. This is (15). The eigenrelation (16) follows by differentiating $\mathbb{E}[e^{sZ_t}] = e^{t\,\psi(s)}$ at $t = 0$. □

Proof of Lemma 5.1. By (16), $(q - \mathscr{L}^{(k)})e^{\gamma z} = (q - \psi_Z^{(k)}(\gamma))e^{\gamma z}$. Taking $\gamma = j$ for each term of (20) and dividing by $q - \psi_Z^{(k)}(j) \neq 0$ (Assumption 5.1(iii)) gives (21). For the homogeneous equation, $e^{\gamma z}$ is a solution iff $\psi_Z^{(k)}(\gamma) = q$; clearing the four Kou denominators gives a degree-six polynomial whose roots are simple and split three-and-three by Assumption 5.1(ii). Boundedness as $z \to +\infty$ excludes $\gamma_{1,2,3}$, and re-centring $e^{\gamma z} = e^{\gamma(z+h_0)}e^{-\gamma h_0}$ absorbs a constant into B_m, giving (22). □

Proof of Proposition 5.2. On $z > -h_0$ write the candidate solution as an exponential sum $F(z) = \sum_\ell a_\ell e^{\beta_\ell z}$, whose exponents are the forcing orders $j = 0,\dots,k$ (coefficients from (21)) and the decaying roots $\gamma_{4,5,6}$ (coefficients $B_m e^{\gamma_{m+3}h_0}$), and set $\widehat{U} = F$ on $z > -h_0$ and $\widehat{U} = 0$ below. The diffusion and upward-jump parts of $\mathscr{L}^{(k)}$ act on $\widehat{U}$ exactly as on F: upward jumps from $z > -h_0$ land above the barrier, and the integrals converge because every β_ℓ lies below $r_{1,+}^{(k)}$ and $r_{2,+}^{(k)}$. The downward-jump part sees the killing. For $z > -h_0$,

$$\int_{-\infty}^{0} \widehat{U}(z+y)\,r\,e^{ry}\,\mathrm{d}y = \int_{-(z+h_0)}^{0} F(z+y)\,r\,e^{ry}\,\mathrm{d}y = \sum_\ell a_\ell \frac{r}{r+\beta_\ell}\left(e^{\beta_\ell z} - e^{-\beta_\ell h_0}\,e^{-r(z+h_0)}\right), \qquad r = r_{j,-}^{(k)},$$

which is valid for every $\beta_\ell \neq -r$ because the range of integration is finite. The first terms reproduce (17), so by Lemma 5.1 $(q - \mathscr{L}^{(k)})\widehat{U}$ equals g_k plus residuals proportional to $e^{-r_{j,-}^{(k)}(z+h_0)}$, $j = 1,2$. These two functions are linearly independent of each other and of g_k under (23), so the resolvent equation holds on $z > -h_0$ if and only if both coefficients vanish: $\sum_\ell a_\ell e^{-\beta_\ell h_0}\, r_{j,-}^{(k)}/(r_{j,-}^{(k)} + \beta_\ell) = 0$, $j = 1,2$. These are rows two and three of the system. Row one is continuity at the barrier, $\sum_\ell a_\ell e^{-\beta_\ell h_0} = 0$, which holds because $\sigma_Z > 0$ makes the barrier regular for the diffusion (Kou and Wang, 2003; Sepp, 2004; Asmussen et al., 2004). Splitting each row into its homogeneous part (where $e^{\gamma_{m+3}(z+h_0)} = 1$ at $z = -h_0$) and its particular part gives $\mathbf{M}_{\mathrm{bar}}^{(k)}\mathbf{B}^{(k)} = -\mathbf{b}^{(k)}$, and Lemma 5.3 gives uniqueness within the exponential ansatz. That this solution is the Laplace transform of the killed expectation follows from the standard verification argument for bounded solutions of the resolvent equation of a killed Lévy process (Kou and Wang, 2003; Asmussen et al., 2004). □

Proof of Lemma 5.3. Subtracting row 1 from rows 2 and 3 replaces entry (i,m), $i = 2,3$, by $-\gamma_{m+3}/(r_{i-1,-}^{(k)} + \gamma_{m+3})$. Removing the two signs and factoring γ_{m+3} from each column,

$$\det \mathbf{M}_{\mathrm{bar}}^{(k)} = \gamma_4\gamma_5\gamma_6\,\det\left(\frac{1}{x_i + \gamma_{j+3}}\right)_{1\le i,j\le 3}, \qquad x_1 := 0,\ x_2 := r_{1,-}^{(k)},\ x_3 := r_{2,-}^{(k)}.$$

By the Cauchy determinant formula,

$$\det\left(\frac{1}{x_i + \gamma_{j+3}}\right) = \frac{\prod_{i<i'}(x_{i'} - x_i)\,\prod_{4\le m<m'\le 6}(\gamma_{m'} - \gamma_m)}{\prod_{i=1}^{3}\prod_{m=4}^{6}(x_i + \gamma_m)},$$

and the $x_1 = 0$ factor $\gamma_4\gamma_5\gamma_6$ cancels the prefactor. Every remaining factor is non-zero: $r_{1,-}^{(k)} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $r_{1,-}^{(k)} \neq r_{2,-}^{(k)}$ by (23) (which is (25) at $k = 0$); the roots are distinct by Assumption 5.1(ii); and $\gamma_m \neq -r_{j,-}^{(k)}$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, where it cannot equal q. □

Proof of Proposition 5.4. Laplace-transforming $\partial_t p_{\text{surv}} = \mathscr{L}^{(0)} p_{\text{surv}}$ with $p_{\text{surv}}(0,\cdot) = 1$ gives $(q - \mathscr{L}^{(0)})\widehat{U}_{\text{surv}} = 1$. Since $\psi_Z^{(0)}(0) = 0$, the particular solution is $1/q$; the homogeneous part is (22) at $k = 0$. The boundary conditions of Proposition 5.2 apply verbatim, with $K_{j,-}^{(0)}[1/q] = 1/q$ because each kernel integrates to one. This gives $\mathbf{M}_{\text{bar}}^{(0)}\mathbf{C} = -\frac{1}{q}(1,1,1)^\top$, uniquely solvable by Lemma 5.3 at $k = 0$. Inverting at $z = 0$ gives p_{surv}. □

Proof of Proposition 7.1. Let $D_t := h_0 + Z_t > 0$ be the distance to the barrier before liquidation. On $\{\tau \le T\}$, almost surely exactly one of three things happens at τ: the diffusion carries Z continuously onto the barrier ($u = 0$, possible because $\sigma_Z > 0$); X_1 jumps down; or X_2 jumps up. Simultaneous jumps have probability zero because N_1 and N_2 are independent. Down-jumps of X_1 arrive at rate $\lambda_1(1-p_1)$ with sizes exponential at rate $r_1 = 1/\eta_{1,-}$, independent of the past; a jump of size y at time $t < \tau$ liquidates exactly when $y > D_{t-}$, with overshoot $u = y - D_{t-}$. By the compensation formula for the Poisson random measure of these jumps, for bounded measurable f,

$$\mathbb{E}\big[f(u)\,\mathbf{1}_{\{\tau\le T,\ X_1 \text{ jump}\}}\big] = \mathbb{E}\int_0^{T\wedge\tau} \lambda_1(1-p_1)\int_{D_{t-}}^{\infty} f(y-D_{t-})\,r_1 e^{-r_1 y}\,\mathrm{d}y\,\mathrm{d}t = P_1\int_0^{\infty} f(u)\,r_1 e^{-r_1 u}\,\mathrm{d}u,$$

with $P_1 := \mathbb{E}\int_0^{T\wedge\tau}\lambda_1(1-p_1)e^{-r_1 D_{t-}}\mathrm{d}t$, after substituting $y = D_{t-} + u$. The inner integral factorises because the exponential density satisfies $r_1 e^{-r_1(D+u)} = e^{-r_1 D}\,r_1 e^{-r_1 u}$. The same argument with rate $r_2 = 1/\eta_{2,+}$ and intensity $\lambda_2 p_2$ applies to up-jumps of X_2, and continuous crossings contribute $P_0 f(0)$. The three cases exhaust $\{\tau \le T\}$, so $P_0 + P_1 + P_2 = p_{\text{liq}}(T)$ and (32) follows. With $f(u) = e^{-ju}$ the right-hand side is $P_0 + P_1 r_1/(r_1 + j) + P_2 r_2/(r_2 + j)$. The matrix with rows $\big(1, r_1/(r_1+j), r_2/(r_2+j)\big)$, $j = 0,1,2$, has determinant proportional to $r_1 - r_2$, so it is nonsingular when $r_1 \neq r_2$ and the weights are determined. □

Replication

All results are produced by the `optimal_long_short` Python package: the resolvent and moment machinery in `moments.py`, `laplace_resolvent.py`, `root_finder.py` and `inversion.py`; exact overshoot recovery in `overshoot_recovery.py`; the overshoot law and expected log wealth in `kelly.py`; the regime resolvent in `regime_switching.py`; the sizing grid in `drift_bayes.py`; stale-aware full-history panels in `full_history.py`; and ECF calibration in the `calibration/` subpackage. Scripts under `jobs/`: `build_merged_4h.py` and `calibrate_btc_eth.py -merged` (headline data and Table 2), `numerical_comparison.py` (Table 3), `mc_convergence.py` (Table 4), `sensitivity_analysis.py` (Figure 1), `overshoot_table.py` (Table 5), `regime_calibration.py` (Table 7), `breakeven_drift.py` (Table 8, Figure 2, the posteriors of Section 9), `kelly_sizing.py` (the validation of Section 7.5), `bootstrap_se.py` (Table 10), `model_comparison.py` (Appendix A.2), and `fetch_universe.py`, `horizon_analysis.py` and `rolling_backtest.py` (Appendix A.3, Tables 11–12; `backtest_bias.py` for the significance of the over-prediction), and `realised_backtest.py` with `fig_realised_backtest.py` (Section 9.3, Table 9, Figure 3). Data are the Aave v3 Ethereum series under `data/AAVE/` (headline window) and `data/AAVE_full/` (full history).

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